DEPARTMENT OF TRANSPORTATION

FEDERAL AVIATION ADMINISTRATION

GREAT LAKES REGION

Fort Worth, Texas

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8R Mid Field RVR and 15L RO RVR access road construction, George Bush Intercontinental, Houston, Texas

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DIVISION 1 - GENERAL REQUIREMENTs
SECTION 1A
SPECIAL CONDITIONS

1A.1 SCOPE.

a. This Specification covers general requirements for construction of RVR. In general, all parts of this Specification covering construction required on project drawings and in other Contract documents are applicable to this Contract.

b. The Contractor shall furnish all plant, labor, materials (except Government-furnished property), equipment, energy, transportation, and other services necessary to construct all elements of the systems required in the Specifications, Drawings, and other Contract documents. Construction shall include all miscellaneous and incidental work necessary for a complete and operational system, whether or not such work is specifically shown or specified.

c. Work for 8R Mid Field RVR and 15L RO RVR access road construction, George Bush Intercontinental, Houston, Texas shall include:

   i. At 8R Mid Point RVR an access road will be constructed from the taxiway CC to the existing RVR sensor. At 15L RO RVR the access road will be extended from the existing access road to the taxiway WP.

1A.2 GOVERNMENT-FURNISHED Material (GFM). Government-furnished Material for this contract is listed on the Government-furnished Material list and is the sole Contract document which validly identifies Government-furnished material under this Contract. The Government-furnished property will be available on site prior to construction.

1A.3 CONTRACTOR-FURNISHED MATERIAL. The Contractor shall furnish all material under this contract per Paragraph 1A.1b, except the Government-furnished material identified on the Government-Furnished Material List. The instruction “by others” on the drawings means equipment that is provided by a third. Contractor will not be required to provide work or material shown as “by others”. The Contractor shall be aware that certain materials to be furnished by the Contractor, may be long-lead-time items. Therefore, the successful bidder should determine the availability of all material immediately after Contract award, and initiate procurement action on long-lead-time items at the earliest possible date. Where the Specifications mention material or equipment by brand, it is regarded as a known acceptable source, as it meets Specifications.

1A.4 SUBMITTALS AND BRAND NAME USAGE.

a. Introduction. Each product required for use in the Contract Drawings and Specifications must meet the actual minimum needs of the Government as demonstrated in the salient characteristics for that product. If a brand name product is used in the Drawings or Specifications, it should be regarded as a "known acceptable source" (i.e., a product that meets the actual minimum needs, and demonstrates the appropriate salient characteristics). The product used can be identical or equal to the brand name product or known acceptable source in meeting the salient characteristics, but it need not exceed the actual minimum requirements. Any brand name product or known acceptable source mentioned will, however, not be required for use in order to

comply with the Specification or Drawing unless those documents make it clear that the brand name product is required, and substitution is prohibited. The following submittal procedure shall be followed in order to:

i. Insure adherence to functional and quality standards in substitution of contractor furnished material.

ii. Provide manufacturers catalog sheets for all material to be furnished and installed by the contractor.

b. Definition. A submittal is a collection of information required by Specifications, or by the Contracting Officer, presenting detailed information on:

i. Material or equipment items the Contractor proposes to use.

ii. Methods or plans of action which the Contractor intends to employ in specific situations.

c. Requirements. All products that a Contractor wishes to use must be approved before use by the Project Engineer or the Contracting Officer's Representative (COR). To gain approval, the Contractor must submit documents and/or samples that demonstrates the product clearly meets the Government's minimum needs, and demonstrates appropriate salient characteristics. All submittals must be in writing. The Contracting Officer shall have the right to require submittals from the Contractor where the Contractor makes an unsolicited change proposal. The information presented in a submittal shall be sufficient to demonstrate that all Specification requirements for the subject material, equipment, methods, or plans, are met by the Contractor's proposal. The informational materials may include documents such as shop drawings, sketches, calculations, data sheets, written plans of action, manufacturers' catalog cuts, brochures, and/or specification sheets. If the Specifications or Contracting Officer requires actual samples of material or equipment, the Contractor shall provide them.

d. Submittal Review. When submitting before the Notice to Proceed date, the Contractor shall send the submittal package(s) directly to the Contracting Officer. When submitting after Contract work has begun, the Contractor shall give submittal packages to the RE, who will forward them promptly to the Contracting Officer or Project Engineer. The Contracting Officer may personally evaluate the submittal, or request Project Engineer to evaluate it. In either case, the submittal will return directly from the Contracting Officer to the Contractor, with the Contracting Officer's approval, approval with comments, or disapproval.

e. Submittal Time Frame. To provide adequate time for document transmission and submittal review, the FAA reserves the right to take two weeks to complete a review, transmission date to transmission date. Terminal navigational aid contracts are brief contracts. The review process can therefore span a substantial portion of the contract period. For this reason:

i. The Contractor is urged to initiate submittals as soon as feasible after contract award, and to expedite document transmission.

ii. The Project Engineer and other reviewers (if any) will expedite reviews and document transmission insofar as feasible. Maximum use of fast document transmission modes (e.g. email, fax, etc.) is encouraged.

f. Procurement Before Approval. The Contractor is advised not to procure any item for which submittal approval is required but not yet granted. If approval is denied, the Contractor will be prevented from installing the disapproved item(s). The Contractor must transmit a new submittal package for the new items replacing the disapproved items, and must procure only approved items. The Contractor shall take responsibility for the delivery and installation of any items installed before submittal approval is granted. The FAA reserves the right to discontinue field work on any item furnished without submittal approval.

1A.5 PRE-CONSTRUCTION CONFERENCE. The Contractor shall attend a pre-construction conference when required by the Contracting Officer or airport management. The Contractor shall abide by all agreements reached at the conference regarding safety practices, ingress and egress routes to the site, maintenance of airport security (locking gates, etc.), deference to air traffic, and other operational procedures.

1A.6 COORDINATION. All coordination between the Contractor and the airport management and local FAA personnel, shall be accomplished through the RE.

1A.7 TEMPORARY ELECTRICAL POWER. Unless otherwise specified, the Contractor shall make all arrangements and pay all costs for temporary electrical power needed for construction of the facility.

1A.8 COMPLIANCE WITH LOCAL AND OTHER CODES. The Contractor shall comply with standards (e.g., National Electrical Code) adopted by the Contract documents, and with local and other codes. Where the requirements of the Specifications and Drawings exceed those of the adopted and local codes, the Contractor shall comply with the requirements of the Specifications and Drawings.

1A.9 SANITARY FACILITIES. Sanitary facilities are not available at the work sites. The Contractor shall provide temporary toilet facilities for their employees as required by OSHA regulations. The locations of the toilet facilities shall be where directed by the RE.

DIVISION 1 – GENERAL REQUIREMENTS
SECTION 1B
SAFETY ON AIRPORTS

1B.2 DEFINITION.

a. Classified Area. For classified area such as the Runway Safety Area (RSA), Object Free Area (OFA), or any other critical areas refer to the plans. Classified areas are specific for every site.

1B.3 GENERAL PRECAUTIONS. The Contractor shall abide by all requirements as specified herein, in the Contract clauses, on the construction safety plan, and as established by airport authorities in the pre-construction conference. The Contractor shall be responsible for thoroughly explaining all safety and security precautions required on the airport to all workers, both under their direct employment and under their subcontractors.

1B.4 CONSTRUCTION WITHIN CLASSIFIED AREAS.

a. Restrictions.

i. Construction within or access through classified areas will not be permitted whenever runways or taxiways defining the classified areas are being used for aircraft operations.

ii. If runways and taxiways within classified areas are required to remain open, construction within the classified areas will be interrupted as necessary to permit normal aircraft operations. No person is allowed in the RSA when runway is in operation.

b. Time Frame. All construction within classified areas shall be completed within the shortest possible time. Construction shall be performed continuously during normal working hours, except as otherwise specified, until all work within the classified areas is completed.

1B.5 CONSTRUCTION WITHIN UNCLASSIFIED AREAS. Construction will be permitted within unclassified areas while aircraft are using adjacent runways and taxiways, excepting as specified elsewhere or established during the pre-construction conference.

1B.6 MAINTENANCE OF AIRCRAFT OPERATING SURFACES. Soil, debris, or loose materials dropped or tracked onto airport roads, runways, taxiways, and ramps shall be immediately swept up and removed. Likewise, all loose material at the job site or dropped enroute to the job site which can be blown onto the above aircraft surfaces, shall be immediately placed in closed containers to prevent damage to aircraft.

1B.7 EQUIPMENT PARKING. All equipment not in use at the close of each day shall be parked in the staging area provided in the construction and safety plan or as directed by the RE.

1B.8 RADIO COMMUNICATIONS. At airports served by airport traffic control towers or airport owner/operator radio communications facilities, (if so directed by the airport management), the Contractor shall furnish and operate two-way radio communications with these facilities when personnel, vehicles, and equipment are required to enter the

aircraft operations area, to obtain proper clearance for construction hazards to aircraft, and at all other times established during the pre-construction conference.

1B.9 TEMPORARY AIRCRAFT PAVEMENT TEMPORARY MARKING AND LIGHTING. These requirements only for part 139 airport and/or when displacing a threshold.

a. Installation. If runway and/or taxiway closure or runway threshold relocation or displacement is required, the Contractor shall install temporary marking or temporary marking and lighting, as shown on the construction safety plan Drawing(s), if any. All temporary marking shall be constructed of plywood, durable fabric, or other approved material, placed and secured so as to pose no threat of damage to aircraft, and which can be easily removed after construction completion.

b. Maintenance. It will be the Contractor's responsibility to maintain the temporary marking and lighting in a condition acceptable to the RE. If marking or lighting is damaged or becomes inoperative, the Contractor shall immediately repair the affected items.

c. Removal. Upon acceptable completion of the work that necessitated runway threshold displacement or runway and/or taxiway closure, the Contractor shall remove all temporary marking and lighting, and shall return the runway and taxiway and lighting configuration to the original condition.

1B.10 ADVISORY CIRCULAR. As a minimum, the Contractor shall adhere to all requirements of Advisory Circular AC 150/5370-2F, "Operational Safety on Airports During Construction" and the included reference documents. This document is available at [https://www.faa.gov/documentLibrary/media/Advisory_Circular/150-5370-2g.pdf]

DIVISION 2 – SITE WORK
SECTION 2A
EARTHWORK AND SITE IMPROVEMENTS

2A.1 DESCRIPTION OF WORK. The extent of earthwork is indicated on the drawings and by the provisions of this section.

2A.2 QUALITY ASSURANCE.

a. Codes and Standards. Perform all earthwork in compliance with applicable requirements of governing authorities having jurisdiction.

b. Testing and Inspection.

i. Soil material, moisture content, and degree of compaction shall conform to ASTM specifications referenced herein. Professional soil testing methods associated with this specification will generally not be required, but the FAA reserves the right to engage a state-licensed soil testing service to resolve disputes regarding adequacy of all earthwork performed.

ii. Visual inspection and qualitative testing shall be performed by the contractor in the presence of, and wherever directed by, the RE.

2A.3 SAFETY REQUIREMENTS.

a. Refer to Division 1 Section 1B for construction within classified and unclassified areas.

b. To protect life, property, and work, all earthwork operations shall be performed in compliance with local and OSHA requirements. The Contractor shall provide all sheeting, shoring, and other bracing as necessary.

c. All trenches in classified areas shall be excavated and backfilled during the same day.

2A.4 JOB CONDITIONS.

a. Existing Utilities.

i. Locate all underground cables, utility lines, and other underground construction before beginning excavation work. Any damage to such lines or construction belonging to the FAA, utility companies, or others, shall be promptly repaired, at contractor's expense, to the complete satisfaction of the owner.

ii. Project drawings generally indicate locations of cables maintained by the Federal Aviation Administration only. The FAA will mark with paint or flags the approximate locations of its own cables. Provide a minimum of 48 hours notice to RE prior to digging so that the cables can be marked.

b. Weather Conditions.

i. Excavating and backfilling for foundations, trenches, and jacking or boring pits, shall not proceed when excessively wet or freezing weather conditions could adversely affect the load-bearing characteristics of the soil, or prevent proper compaction.

ii. When freezing weather is expected, excavations shall not be made to full depth unless concrete or conduits can be placed immediately. If an excavation is already at full depth, the excavation shall be protected from frost.

c. Drainage.

i. All excavations shall be continually drained by natural means or pumping to prevent any decrease in soil bearing capacity or damage to poured foundations or to trenches.

ii. Provide and maintain pumps, well points, sumps, suction and discharge lines, and other dewatering system components necessary to convey water away from excavations.

iii. Establish and maintain temporary drainage ditches and other diversions outside excavations limits to convey rain water and water removed from excavations to collecting or run-off areas. Do not use trench excavations as temporary drainage ditches.

2A.5 MATERIALS.

a. Structure and Foundation Materials. In-place undisturbed inorganic soils will be adequate to support all project structures, unless otherwise indicated. Highly organic soils (topsoil, peat, and swamp location soils) shall be removed entirely from areas to be occupied by structures.

b. Backfill and Fill. Material shall be inorganic soil excavated from site, or borrow comprised of inorganic soil approved by the RE. All such soils shall be free of rock, gravel, and cohesive lumps greater than two inches in any direction, and debris, waste, vegetation, frozen material, and other deleterious materials.

c. Base Course for Concrete Slabs. Material shall be a graded mixture of washed crushed stone or crushed or uncrushed gravel with 100% passing a 1 ½ inch sieve, and not more than 5% passing a number 4 sieve.

2A.6 SITE PREPARATION.

a. Clearing and Grubbing. The Contractor shall scalp areas where excavation or embankment will be made. Scalping shall include the removal of materials such as trees, brush, roots, sod, grass, residue of agriculture crops, sawdust, and decayed vegetable matter, from the surface of the ground. These materials shall be removed from the site and disposed of off airport property.

b. Topsoil Removal.

i. Topsoil shall be considered soil containing visible vegetable matter and black loam that will not compact with the usual compacting methods.

ii. Unless otherwise specified, topsoil shall be removed from all areas to receive fill, granular surfacing, pavement, and structures, and from all areas where subsoil excavating is required, such as for roadway cuts and ditches. Dispose of excess topsoil on or off airport property, as directed by the RE, at no additional cost to the Government.

2A.7 EXCAVATION.

a. Excavation Classification. Excavation is unclassified and includes excavation to subgrade elevation indicated, regardless of character of materials and obstructions encountered excepting as qualified herein.

b. Rock Excavation. If rock is encountered above the design footing elevations of any facility structure, such foundation shall bear entirely on clean solid rock or on soil, but not on both. If the soil-and-rock bearing condition is encountered, the RE will determine which material shall support the structure. If rock surface is used, it shall be reasonably level or shall be stepped to make level segments.

c. Unauthorized Excavation. Removal of materials beyond design subgrade elevations or dimensions without specific direction from the RE constitutes unauthorized excavation. Remedial work for such excess excavation shall be as directed by the RE at the Contractor's expense.

d. Additional Excavation. When any excavation has reached required subgrade elevation, notify the RE, who will inspect soil conditions. If the RE determines that the soil possesses inadequate bearing capacity, carry such excavation deeper as directed by the RE.

e. Excavation for Structures.

i. Conform to elevations and dimensions shown within a tolerance of plus or minus 0.10 foot, and extending a sufficient distance from footings and foundations to permit placing and removal of concrete formwork, installation of services and other construction, and for inspection.

ii. In excavating for footings and foundations, remove loose soil and compact bottom of hole prior to placing reinforcement and concrete.

f. Excavation for Cable and Conduit Trenches.

i. Excavate in compliance with lines and depths shown on drawings. Minimum trench depth shall allow for cables to be installed at the depths specified in Paragraph 16F.3.b.ii., unless otherwise specified. Slope trenches to same elevations as conduits where cables will be routed to a building interior. Minimum trench width shall be that required to accept power-operated mechanical tampers.

ii. Grade bottom surfaces of trenches to provide uniform bearing and continuous support for cable and conduit.

iii. Material excavated in excess by error, or due to unsuitable bearing, shall be replaced with mechanically compacted inorganic soil.

iv. If solid rock is encountered, the RE will decide if such rock need be removed or if an alternate trench route or lesser depth conduit installation will be acceptable.

v. If a trench must cross a concrete or asphalt paved surface, all cuts shall be saw cuts, unless otherwise specified.

2A.8 COMPACTION.

a. General.

i. All compaction shall be accomplished by using power-operated mechanical equipment except for limited use of manual tampers in constricted areas. Operate all power equipment as herein specified to achieve the minimum degree of compaction subject to acceptance by testing.

ii. Cohesive soils are defined herein as those containing less than 60 percent sand, gravel, or stone. Percentages greater than 60 percent are herein termed non-cohesive soils.

b. Cohesive Soil Compaction.

i. Use sheepsfoot roller of such minimum weight that at least 200 psi will be transmitted to surface area of studs or feet. Operate at speeds not exceeding 4 mph on each layer of fill until roller walks itself to top of grade.

ii. Use motor-operated soil tamper (stomper) in confined areas, including trenches, on each layer of fill until no further visible consolidation is evident.

iii. Use a heavy blunt tamping rod on each layer of fill in the most constricted locations where power equipment cannot be used.

c. Non-Cohesive Soil Compaction.

i. Use pneumatic tire roller fully loaded and weighing not less than 275 pounds per inch of tire tread width. Operate at speeds not exceeding 4 mph. A minimum of ten passes of the roller is required on each layer of fill.

ii. Use motor-operated vibratory tamper in confined areas, including trenches, on each layer of fill until no further visible consolidation is evident.

iii. Use heavy blunt tamping rods on each fill layer in constricted locations where power equipment cannot be used.

d. Moisture Control.

i. Where soil material must be moisture-conditioned before compaction, uniformly apply water to a layer of soil material in such quantity that free water will not appear on the surface during or subsequent to compaction operations.

ii. Scarify and air-dry soil material that is too wet to permit compaction to specified density.

e. Percentage of Maximum Density Requirements.

i. General Requirements. The required densities for cohesive and non-cohesive soils are determined by quantitative testing procedures defined by ASTM Standards D 1557 and D 4253/4254, respectively. To assure compliance, the Contractor may arrange for such professional soil testing services, at no additional cost to the Government. The FAA, at its expense, may also make such arrangements if qualitative testing procedures appear inadequate.

ii. Structures, Slabs, and Access Roads/Parking Areas. Compact top surfaces of subgrade and each layer of backfill or fill material to 90% of maximum density for cohesive soils, or to 95% relative density for non-cohesive material.

iii. Turf and Non-Vehicular Surfaced Areas. Compact top surfaces of subgrade and each layer of backfill or fill material to 90 percent of maximum density for cohesive soils, or to 90 percent relative density for non-cohesive material.

f. Qualitative Testing and Inspection Procedure.

i. General. The Contractor shall perform qualitative soil compaction testing and inspection procedures for each type of backfill or fill material used wherever directed by, and in the presence of, the RE. Special attention shall be given to the backfilling of structures and trenches.

ii. Qualitative Testing.

1. Qualitative soil testing will consist of comparing the resistance to penetration of undisturbed soil to that of compacted backfill of the same composition. For borrow material the penetration comparison shall be made between maximum test sample density and in-place fill density.

2. A soil penetration device (penetrometer) indicating depth and force exerted shall be utilized. Compaction will be adequate if backfill or fill possesses at least 95% of the resistance to penetration of undisturbed soil or test sample, respectively.

3. Borrow test sample shall be a four-inch-deep (compacted measurement) layers of soil, aerated or moistened as directed by the RE, and compacted by power equipment until no further consolidation occurs, as approved by the RE.

iii. Concrete Slab Base Course. Compact with vibratory tamper until no further visible consolidation is evident.

2A.9 BACKFILL AND FILL.

a. Structure Foundations. Backfill or fill as promptly as work permits, but not until completion of the following:

i. Acceptance of construction below grade.

ii. Recording locations of underground conduit.

iii. Removal of concrete formwork, bracing, trash, and debris.

b. Ground Surface Preparation. Remove vegetation, debris, topsoil, and unsatisfactory subsoil from ground surface, and compact the subgrade, prior to placement of fill layers.

c. Placement and Compaction.

i. Place acceptable backfill and fill materials in layers not more than eight inches in loose depth for material to be compacted by heavy equipment, and not more than four inches in loose depth for material to be compacted by hand-operated tampers.

ii. Before compaction, moisten or aerate each layer as necessary to provide optimum moisture content. Do not place backfill or fill on water, ice, snow, frozen soil, or excessively wet soil.

d. Cable Trench Backfill.

i. Before laying cables, inspect the bottom of the cable trench. If it is not smooth, or if any rock or stone that would be retained on a 1/4-inch sieve is present, place a two-inch layer of bedding material, according to Paragraph (2) below, in the trench. Do not compact this layer. Lay cables on top of this layer.

ii. The first layer of backfill material over cables shall be three inches deep, loose measurement, and shall be sand or other homogeneous inorganic soil containing no mineral aggregate particles that would be retained on a 1/4-inch sieve. This layer shall not be mechanically compacted.

iii. The second layer, in turf and crushed rock surface areas, shall be four inches deep, loose measurement, and shall contain no mineral aggregate particles that would be retained on a one-inch sieve. Subsequent layers shall be clean soil containing no rock particles larger than two inches in their largest dimension.

iv. Except for surfacing material, all layers of trench backfill, for areas to be paved or surfaced with crushed rock, shall be sand, placed and compacted as required for access roads.

1. If a trench crosses an area surfaced with crushed rock, the top 12 inches of trench backfill shall be crushed rock, placed and compacted as required for access roads. The finished grade elevation of the crushed rock backfill shall equal the grade elevation of existing adjacent crushed rock.

2. If a trench crosses an area surfaced with concrete or asphalt pavement, the pavement shall be replaced with materials of the same composition, thickness, and degree of compaction as the adjacent pavement structure, except that the crushed rock base shall be a minimum of 12 inches deep. Replacement concrete shall have a 28-day compressive strength of 3,000 psi. Finished grade of the pavement patch shall be flush with the adjacent pavement surfaces.

e. Backfill and Fill Surface Elevations. Finished grade, shown on the drawings, is the top surface of turf and crushed rock or crushed stone surfaced areas. Therefore, make allowances for six inches of topsoil and depths as detailed or specified for surfaced areas when establishing top surface of fill or backfill.

2A.10 GRADING.

a. General. Uniformly grade areas within limits of grading, including adjacent transition area. Smooth the finished surfaces within specified tolerances, and compact with uniform slopes between points where elevations are indicated, or between such points and existing grades.

b. Grading Outside Building Lines. Grade areas adjacent to building lines to drain away from structures and to prevent ponding. Finish areas to receive topsoil and surfacing within 0.10 feet above or below required subgrade elevations.

c. Grading Surface of Fill Under Building Slabs. Grade smooth and level and to proper elevation to within a tolerance of 1/2 inch when tested with a 10-foot straightedge.

2A.11 MAINTENANCE.

a. Protection of Graded Areas. Protect newly graded areas from traffic and erosion. Keep free of trash and debris.

b. Reconditioning. Where compacted areas are disturbed by construction operations, adverse weather, or where any settlement has occurred, scarify surface, add acceptable fill, reshape, grade, and compact as necessary.

2A.12 DISPOSAL OF EXCESS AND WASTE MATERIALS. Remove and dispose of all excess soil and waste material from the project site and adjacent lands. All costs associated with disposal shall be at Contractor's expense.

DIVISION 2 – SITE WORK
SECTION 2B
CRUSHED AGGREGATE ROAD AND SITE SURFACING

2B.1 DESCRIPTION OF WORK. The extent of work is indicated on the drawings and by the provisions of this section.

2B.2 STATE SPECIFICATIONS. State highway construction specifications, latest edition, form a part of this Specification and are applicable for all work unless otherwise specified. This referenced specification will be hereinafter referred to as "State Specifications." Disregard all references in the State Specifications to layout of work by others, and to measurements and payments. All layout work will be accomplished by the Contractor, and payment for all work under this section will be a part of the lump-sum contract.

2B.3 MATERIALS.

a. Geotextile.

i. Application. The most common application of geotextiles in FAA navaids construction is as a separator. In this application, the geotextile is placed over prepared roadway subgrade soil, and crushed aggregate is placed and compacted on top of the geotextile. The geotextile permits water to permeate into the subgrade, while preventing the aggregate from mixing with the subgrade soil. The geotextile specified below is for application as a separator.

ii. Separator Geotextile Selection Criteria. The geotextile fibers, and the threads used in joining the geotextile by sewing, shall consist of long chain polymeric fibers composed of polypropylene, polyester, polyolefins, or polyamide. Both the geotextile and threads shall be resistant to chemical attack, mildew, and rot. The geotextile shall conform to the physical property requirements listed in the following table. All values shall represent certifiable minimum values in the weakest principle direction of the fabric.

Property	Test Method Requirements		
Tensile Strength	ASTM	D-4595	2460 lbs/ft
Tensile Strength (at 2% strain)	ASTM	D-4595	480 lbs/ft
Tensile Strength (5%)	ASTM	D-4595	1212 lbs/ft
Tensile Strength (10%)	ASTM	D-4595	2340 lbs/ft
Factory Seam Strength	ASTM	D-4632	1250 lbs/ft
Flow Rate	ASTM	D-4491	50 gpm/ft^2

Permeability	ASTM	D-4491	0.04 cm/sec
Permittivity	ASTM	D-4491	0.6 sec^-1
Apparent Opening Size	ASTM	D-4751	U.S. Sieve #30, max
UV Resistance	ASTM	D-4355	80% str. retained

iii. Geotextile Fabric Width. Fabric width shall be at least 12 feet for the normal 12-foot-wide access road. Fabric in other vehicular areas shall be cut to fit, and overlapped per Paragraph 2B.4c(2)(b), below, to fully cover such areas.

b. Landscape Fabric.

i. Application. Landscape fabric shall be applied under all non-roadway crushed rock surfacing. In these applications, landscape fabric is placed over prepared walkway subgrade soil, and crushed aggregate is placed and compacted on top of the landscape fabric. The Contractor shall have the option of substituting geotextile per Paragraph 2B.3a, for landscape fabric.

ii. Landscape Fabric Selection Criteria. The landscape fabric fibers shall be a needle punched nonwoven geotextile composed of polypropylene, polyester, polyolefin, or polyamide fibers. The fabric shall be resistant to chemical attack, mildew, and rot. The fabric shall conform to the physical property requirements listed in the following table. All values shall represent certifiable minimum values in the weakest principle direction of the fabric.

Property	Test Method Requirements		
Grab Tensile Strength	ASTM	D-4632	120 lbs
Grab Elongation	ASTM	D-4632	50%
Trapezoidal Tear	ASTM	D-4533	50 lbs
CBR Puncture Strength	ASTM	D-6241	310 lbs
Apparent Opening Size	ASTM	D-4751	U.S. Sieve #70, max
Permittivity	ASTM	D-4491	1.7 sec^-1
Flow Rate	ASTM	D-4491	1.35 gpm/ft^2

iii. Landscape Fabric Width. Fabric width shall be 3 feet wide for a 3-foot-wide crushed rock walkway. Fabric in other walkway areas shall be cut to fit, and overlapped per Paragraph 2B.4.c.ii.2., below, to fully cover such areas.

c. Crushed Aggregate Surfacing.

i. Crushed rock or crushed stone aggregate shall comply with State Specification quality requirements for crushed rock or crushed stone used for road surface course,

and shall be of the State gradation most closely conforming with the following gradation:

Sieve Size	Total Passing, Percent
1-inch	100
$\frac{3}{4}$-inch	80-100
$\frac{3}{8}$-inch	30-60
No. 4	48-65
No. 8	35-50
No. 30	19-30
No. 50	13-23
No. 100	7-15
No. 200	0-8

ii. A certified sieve analysis, referenced to State Specification gradation, shall be submitted to the RE for approval.

2B.4 Construction.

a. General Requirements. All earthwork requirements in Section 2A for areas to receive surfacing are applicable, excepting as qualified herein. Where the additional work or more stringent requirements in this section conflict with Section 2A, requirements herein shall prevail.

b. Foundation Preparation.

i. Foundation Material. All topsoil shall be removed from areas to receive paving and surfacing or fill under such surfaces. Only inorganic soil shall exist under surfaced or paved areas.

ii. Compaction. Compact as required in Section 2A.

iii. Grading. Shape with motor grader to achieve such surface trueness that when tested with a 10-foot straightedge, no deviation greater than 1/2-inch shall exist.

iv. Corrective Work. Any ruts or soft-yielding spots that may appear in the subgrade, any areas having inadequate compaction, and deviations of the surface from the requirements specified shall be corrected by loosening, removing, and adding approved material and reshaping and recompacting the affected areas to line and grade, and to the specified density.

c. Geotextile or Landscape Fabric.

i. General. Geotextile or landscape fabric, if required on the drawings, shall be installed on prepared subgrade for all areas that will experience vehicular traffic or pedestrian traffic, respectively.

ii. Construction Requirements.

1. Prepared subgrade and foundations shall be compacted smooth and level as specified elsewhere and as shown on the drawings.

2. The fabric shall be rolled out directly upon the prepared surface, and shall not be dragged over any surface. Fabric in place shall have a smooth surface and shall be free of folds, wrinkles, cuts, or other imperfections. Individual panels of fabric shall be overlapped at least 24 inches, with the preceding layer overlapping the following layer in the direction that surfacing material will be spread. No vehicular traffic will be permitted directly upon the fabric.

d. Crushed Aggregate Surfaced Areas and Crushed Aggregate Base Course for Bituminous Pavement.

i. Spreading. Crushed aggregate surfaced areas and base course shall be constructed in one or more layers of maximum 6-inch compacted thickness each. Crushed aggregate shall be deposited directly and uniformly on the prepared subgrade, if no geotextile or landscape fabric is used. If geotextile fabric is required, the aggregate shall be back-dumped on the fabric, and machine spread in the direction of overlap. Dumping in windrows, which requires excessive rehandling, will not be permitted. When deposited, the aggregate shall be free from segregation, and shall require minimum blading or manipulation.

ii. Compaction and Grading.

1. Each layer of aggregate shall be compacted using equipment required in the State Specifications. For compacting aggregate on a geotextile or landscape fabric, use a smooth-drum roller. Compaction shall closely follow the spreading operation to prevent loss of contained moisture or displacement of materials.

2. When the surface stability of the crushed aggregate cannot be obtained due to lack of fines, additional fines shall be added to the upper portion of the course in an amount sufficient to secure stability, at no additional cost to the Government. In no case, however, shall the quantity of fines added increase the percent passing the Number 200 sieve by more than 15 percent in the upper portion.

3. Any irregularities or depressions that develop in the layers under rolling operations shall be corrected by loosening the material and removing or adding aggregate and rerolling. The rolling shall be continued until the surface is shown to be smooth and uniform, and to such trueness that when tested with a 10-foot straightedge it shall not show any deviation in excess of 1/4-inch. At all places not accessible to the roller, the aggregate of each layer shall be tamped separately and compacted to grade and line with mechanical tampers.

4. If any subgrade material is worked into the aggregate material during the compacting or finishing operations, all granular material within the affected

areas shall be removed and replaced with new aggregate. The RE may restrict hauling or traffic over the completed or partially completed base after inclement weather or at any time when the subgrade is soft, and there is a tendency for the subgrade material to work into the base material.

5. If considered necessary by the RE, water shall be applied to each layer to aid in compaction and prevent segregation of the material. Disc or harrow surfacing material during moistening operations to secure uniform moisture distribution. Add water in a manner that will not soften the subgrade. All work associated with the additional water shall be accomplished at no additional cost to the Government.

6. The aggregate shall be compacted to 95 percent maximum density as determined by AASHTO-T99. Compaction shall continue until no further discernible compaction is evidenced under action of the compaction equipment. If in the opinion of the RE, the required degree of compaction has not been achieved, testing in accordance with the standard will be conducted and paid for by the Contractor. If testing confirms unacceptable compaction, reconstruction or other remedial work may be required by the Contractor at no additional cost to the Government.

DIVISION 2 – SITEWORK
SECTION 2D
TOPSOIL AND GRASS COVER

2D.1 DESCRIPTION OF WORK. The extent of topsoil placement and establishment of grass cover is indicated on the drawings and the provisions of this section.

2D.2 GENERAL REQUIREMENTS. All areas of the project site, access road right-of-way, and cable trench routes, which will not be occupied by pavement, crushed rock/stone surfacing, or other construction, shall receive preparation and grass seed planting and maintenance. Included in this work will be off-site turf reconditioning and replacement for those areas damaged by construction operations.

2D.3 MATERIALS.

a. Topsoil. Material shall be that removed from project site location preparatory to trenching and site construction. Reuse only that part of stockpiled topsoil reasonably free of subsoil, trash, roots, stumps, weeds, debris, litter, and stones larger than 2 inches.

b. Fertilizer. Provide complete fertilizer of 5-10-5 composition (percentages of nitrogen, phosphorous, and potash, respectively).

c. Grass Seed. Provide fresh, clean, new-crop seed complying with tolerance for purity and germination established by Official Seed Analysts of North America. Provide the following seed mixture composed of grass species and proportions:

Proportion by weight	Common Name
45%	Kentucky Bluegrass
5%	Perennial Ryegrass
35%	Redtop
15%	White Clover

d. Anti-Erosion Mulch. Provide clear, seed-free salt hay or threshed straw of wheat, rye, oats, or barley. Anchor the mulch sufficiently to prevent it from being blown away.

2D.4 PREPARATION FOR PLANTING.

a. Subsurface Preparation. After completion of all construction operations that could disturb topsoil areas, subgrades shall be cleared free of waste and stones larger than 2 inches, then tilled to a depth of 3 inches, and graded to remove surface irregularities.

b. Topsoil and Surface Preparation. Spread topsoil uniformly to provide a 6-inch layer, after compaction, on all fill and backfilled areas to receive grass seed. Compact with a roller weighing 85-100 pounds per foot of width. Subgrade and topsoil shall be damp when work is performed, but not wet, dusty, or frozen.

c. Preparation of Unchanged Grades. Where seed will be planted in areas that have not been altered by grading, prepare soil for planting as follows: till to a depth of 6 inches, apply fertilizer, remove high areas and fill depressions, till soil to a homogenous mixture of fine texture, free of lumps, stones, roots, and trash, and compact as above.

d. Fertilizer. Apply fertilizer at the rate of 30 pounds per 1000 square feet of area. Mix fertilizer into top 2 inches of topsoil.

e. Surface Preparation. Fine grade to a smooth, even surface and to a loose, uniformly fine texture. Roll, rake, and drag seeded areas, remove ridges, and fill depressions as necessary to meet finish grades. Limit work to areas that can be planted immediately.

Moisten prepared areas before planting if soil is dry. Allow surface mixture to dry and proceed with seeding.

2D.5 SEEDING.

a. Seed Condition. Do not use old, wet, or moldy seed. The seed shall be dated with the year in which it is being applied.

b. Sowing. Use spreader or sowing machine. Do not seed when wind velocity exceeds 5 mph. Distribute seed evenly over entire area by sowing equal quantities in two directions at right angles to each other. Seed at rate of 5 pounds of seed per 1000 square feet of area. Rake seed lightly into top 1/8 inch of soil, roll lightly, and water with a fine spray.

c. Protection. Spread mulch uniformly to form a continuous loose blanket after completion of seeding operations. Anchor the mulch sufficiently to prevent it from being blown away.

2D.6 MAINTENANCE. Maintain for a period of 60 days after seeding. If maintenance for a 60-day period is not feasible, the Contractor shall use a commercially available mix of seed, fertilizer, and long-lasting mulch, or a preseeded anchored mat that can be maintained less frequently with the same results. Water, fertilize, regrade, and replant as required to establish smooth, acceptable turf.

DIVISION 2 - SITE WORK
SECTION 2E
MISCELLANEOUS SITE IMPROVEMENTS

2E.1 DESCRIPTION OF WORK. The extent of miscellaneous site work is indicated on the drawings and by the provisions of this section.

2E.2 REMOVAL OF EXISTING FOUNDATIONS. All foundations of removed buildings, trailers, antenna supports, or other structures shall be removed to a minimum depth of two feet below final grade and backfilled with compacted earth in accordance with Section 2A, unless noted otherwise on the site drawings. The site areas shall be graded smooth and topsoil added to match the original terrain, unless otherwise specified.

2E.3 REPLACEMENT OF SURFACING AGGREGATE AND PAVEMENT. For replacement of surfacing aggregate and pavement removed for trenching operations, see Paragraph 2A.9.d.iv. above.

2E.4 SPECIAL SURFACING

a. General. If required on the drawings, surfacing for MALSR stations, and all other small surfacing areas within 300 feet of runway and taxiway edges, shall be the material placed as required below. Special surfacing requirements will preclude displacement onto aircraft operating surfaces.

b. Material and Installation. Crushed rock or crushed stone surfacing at light unit locations shall be 6 inches deep and centered on the units. Material shall be 1 1/2 to 2 inches nominal size washed crushed rock. No substitutions will be accepted. Rock shall be tamped as tightly as material permits. Finished surface shall be flush with existing surrounding grade.

DIVISION 3 – CONCRETE
SECTION 3A
CONCRETE FORMWORK AND REINFORCEMENT

3A.1 DESCRIPTION OF WORK. Extent of work is indicated on the drawings and by the requirements of this section.

3A.2 CONCRETE FORMWORK

a. Design of Forms. Forms shall conform to shapes, lines, and dimensions of the members shown on the plans, and shall be sufficiently tight to prevent leakage of mortar. They shall be properly tied together so as to maintain position and shape.

b. Form Removal. Forms shall not be loosened or removed until the concrete members have acquired strength sufficient to support their own weight. No additional loads shall be placed on the concrete for at least 48 hours after placing.

c. Form Ties. Form ties for concrete shall be of a type that will break back 1 1/2 inches from the concrete surface. Ties shall be removed to a minimum depth of 1 1/2 inches, and the surface patched.

3A.3 CONCRETE REINFORCEMENT.

a. Materials. Reinforcement bars shall conform to "Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement", ASTM A-615. All bars shall be Grade 60 deformed bars.

b. Cleaning and Bending Reinforcement. At the time concrete is placed, metal reinforcement shall be free from rust scale or other coatings that will destroy or reduce the bond. All bent bars shall be bent cold. No bars partially embedded in concrete shall be field bent except as shown on plans.

c. Placing Reinforcement. Metal reinforcement shall be accurately placed according to the plans, and adequately secured in position by concrete, metal, or other approved chairs, spacers, or ties.

d. Splices in Reinforcement. No splices or reinforcement shall be made except as shown on the plans or as authorized by the RE.

e. Concrete Protection for Reinforcement. The reinforcement shall be protected by the thickness of concrete shown on the drawings. Where not shown, the thickness of concrete over the reinforcement shall be as follows:

i. Where concrete is deposited against the ground without the use of forms, not less than 3 inches.

ii. Where concrete is exposed to the weather or to the ground but placed in forms, not less than 2 inches for bars larger than number 5, and 1 1/2 inches for number 5 bars or smaller.

DIVISION 3 – CONCRETE
SECTION 3B
CAST-IN-PLACE CONCRETE

3B.1 DESCRIPTION OF WORK. The extent of work is indicated on the drawings and by the provisions of this section.

3B.2 MATERIALS. Cement shall conform to Specification for Portland Cement ASTM C-150, Type I, or Specification for Air-Entraining Portland Cement ASTM C-175, Type 1A, unless otherwise specified. The concrete shall have a minimum 28-day compressive strength of 3,000 PSI, a maximum slump of 4 inches, and a maximum aggregate size of 1-inch. The concrete mix shall contain an air-entraining admixture. Air content shall be 5 to 7 percent. The Contractor shall give the RE a certificate from the concrete supplier, bearing the intended job mix and certifying that the concrete delivered will meet the above requirements. The Contractor shall obtain approval of the job mix from the RE prior to placing concrete.

a. Water Reducing Admixtures (WRA). (ATSM D-494)

i. If Water Reducing Admixtures (WRA) are used, the slump may be increased to up to 6 inches as long as the concrete maintains greater than or equal to 3,000 PSI compressive strength, not have aggregate greater than 1 inch, and maintain between 5-7% air entrainment.

ii. Water may not be withheld or added on site to increase slump further if WRAs are used.

3B.3 PREPARATION OF EQUIPMENT AND PLACE OF DEPOSIT.

a. Before placement, all equipment for mixing and transporting the concrete shall be cleaned. All debris and ice shall be removed from the places to be occupied by the concrete. Forms shall be thoroughly wetted (except in freezing weather) and oiled prior to placing reinforcing steel. The reinforcement shall be thoroughly cleaned of ice, dirt, rust scale, or other coatings.

b. Water shall be removed from place of deposit before concrete is placed. All laitance and other unsound material shall be removed from hardened concrete before additional concrete is added.

3B.4 CONVEYANCE. Concrete shall be conveyed from the mixer to the place of final deposit by methods that will prevent segregation or loss of materials. Equipment for chuting concrete shall be of such size and design so as to ensure a continuous flow of concrete at the delivery end without segregation of materials.

3B.5 PLACEMENT.

a. Concrete shall be placed within 1 1/2 hours after mixing begins. Concrete shall be deposited as nearly as practicable in its final position to avoid segregation due to rehandling or flowing. The placing of concrete shall be carried on at such rate that concrete is at all times plastic, and flows readily into the spaces between the bars. No concrete that has been contaminated by foreign material shall be used, nor shall retempered concrete be used.

b. When placing is started, it shall be carried on as a continuous operation until placement is completed.

i. Placement shall be done monolithically unless otherwise annotated on the prints.

c. Concrete shall be placed in layers not exceeding 18 inches deep, and vibrated in place. During and immediately after depositing, the concrete shall be consolidated by vibrators. The concrete shall be thoroughly worked around reinforcement, around embedded fixtures, and into corners. Accumulations of water on the surface of the concrete due to water gain, segregation, or other causes, shall be prevented as much as possible by employing proper placement, consolidation, and finishing practices.

Provisions shall be made to remove such water as may accumulate, so that under no conditions will concrete be placed in such accumulations.

d. Vibrators shall be the internal immersion type, operating at speeds of not less than 7,000 RPM. Vibrators shall be kept constantly moving in the concrete and shall be applied at points uniformly spaced not further apart than the radius over which the vibrator is visibly effective. The entire depth of a new layer of concrete shall be vibrated. The vibrators shall penetrate several inches into the layer below to insure thorough union of the layers. The vibrator shall not be held in one location long enough to draw a pool of grout from the surrounding concrete. Vibration shall be such that the concrete becomes uniformly plastic.

3B.6 FOOTINGS. All footings and foundations without footings shall bear on firm, undisturbed soil.

a. Top two feet of cylindrical piers shall be formed in a fiber form and finished flat within the confines of the fiber forms. No spillage (mushrooming) over the tops of forms will be permitted. Where conduit emerges from vertical surfaces of concrete piers, no appreciable amount of concrete shall be permitted to spill through forms adjacent to such conduit.

b. Fiber forms for cylindrical concrete piers shall be spirally constructed of laminated plies of fiber. The total wall thickness shall be as published by the manufacturer. The width of each ply shall not be less than 6 inches. Plies shall be laminated with an adhesive of a non-water-sensitive type, with a proven record of satisfactory service in concrete forms. The exterior surface shall be uniformly wax impregnated for weather and moisture protection. The interior surface shall be coated with pure polyethylene uncontaminated by paraffin or other additives. A-Coated Sonotube forms by Sonoco Products Company of Hartsville, South Carolina, are among the products that meet these Specifications.

c. Remove all loose soil from bore holes so that concrete will bear on undisturbed soil.

3B.7 ANCHOR BOLT INSERTS. No drilling for or placing of anchor bolt inserts or anchors will be permitted in concrete for a period of three days after placement, unless noted otherwise on the drawings.

3B.8 CURING.

a. Provision shall be made for maintaining concrete in a moist condition for a period of at least 5 days after placement.

b. In lieu of wet curing, one coat of a concrete curing sealer which forms a film over the concrete surface, may be used for curing the concrete. The sealer shall meet the ASTM C-309 standard for moisture retention. The compound shall be a type that does not permanently discolor the concrete. Dayton Superior J11W is one of the products which meet this Specification. Where concrete is to receive vinyl composition tile (VCT), or any other flooring, curing compound shall not be used, and only wet sheet curing is allowed. On exposed surfaces, application shall be made immediately after the concrete has been finished. If there is any delay, the concrete shall be kept moist until the application is made. After the forms are removed, the concrete shall be sprayed lightly with water, and then the coat of curing compound applied. If the forms (wood only) cannot be removed within 48 hours, they shall be wetted down and kept wet until their removal, and then the compound applied as above.

3B.9 COLD-WEATHER REQUIREMENTS.

a. Adequate equipment shall be provided for heating concrete materials and protecting concrete during freezing or near-freezing weather. No frozen materials or materials containing snow or ice shall be used. Concrete shall not be placed on frozen soil.

b. All reinforcement, forms, fillers, and ground which will make contact with concrete shall be free from snow and ice. Whenever the temperature of the surrounding air is below 40°F, all concrete placed in forms shall have a temperature of 45°F or higher, after placement. Adequate means shall be provided for maintaining this temperature for 4 days. Any additional time necessary to ensure proper curing of the concrete shall be provided as directed by the RE. The housing, covering, or other protection used in connection with curing, shall remain in place and intact at least 24 hours after the

artificial heating is disconnected. Do not use salt or other chemicals to prevent freezing.

3B.10 HOT-WEATHER REQUIREMENTS.

a. In hot weather, suitable precautions shall be taken to avoid drying of the concrete prior to finishing operations. Use of windbreaks, sunshades, fog sprays, or other devices shall be provided as directed by the RE.

b. Concrete deposited in hot weather shall not have a placing temperature that will cause difficulty from loss of slump, flash set, or cold joints. Concrete temperature shall be less than 90°F.

3B.11 DELIVERY TICKETS. At the time of concrete delivery, the Contractor shall give the RE a copy of the delivery ticket bearing the quantity, strength, and air entrainment of the concrete delivered.

3B.12 CONCRETE TESTS. The Contractor shall submit a concrete testing company and gain approval prior to the Notice to Proceed (NTP). Concrete testing will be performed at the Contractor's expense, at the direction of the Contracting Officer Representative. Concrete testing will not exceed once per 50 cubic yards or once per day (whichever is greater), unless concrete tests do not meet the specification. These tests shall include, temperature, air entrainment, slump, and concrete cylinders for strength.

3B.13 SLUMP. Concrete shall be tested for consistency at the mixer or at the place of deposit if delivered ready-mixed. The sample shall be taken immediately from the batch and tested by the Contractor in the presence of the RE in accordance with ASTM C143. Concrete with slump in excess of 4 inches shall be rejected, unless water reducing admixtures were used, then slump can be up to 6 inches.

3B.14 Air Entrainment. Air content shall be determined by ASTM C231, "Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method". When necessary to increase air content, additional air-entraining admixture identical to that already in the mixture shall be added to the mixer. The contractor shall make all arrangements for air content tests when requested by the Contracting Office Representative. Tests shall be performed in the presence of the Contracting Officer Representative. The Contractor will pay for all costs of the tests. Concrete with air contents outside the 5 to 7 percent range shall be cause for rejection.

3B.15 Strength. Four concrete cylinders are to be made in the dimensions of 6 inch diameter and 12 inches in height or 4 inch diameter and 8 inches in height. The first one shall be broken at 7 days, the next at 14 days, and the final 2 shall be broken at 28 days. If concrete specified in the mix design has a design strength of over 8,000 PSI, then 4 inch diameter by 8 inches in height cylinders will be recommended.

DIVISION 5 – METALS
SECTION 5A
MISCELLANEOUS METALS

5A.1 DESCRIPTION OF WORK. Extent of metal work is indicated on the drawings and by the provisions of this section.

5A.2 MATERIALS.

a. Structural Steel Shapes and Plates. ASTM A992 for W-shapes, and ASTM A36 for S, C, MC, and L-shapes, as well as steel plates.

b. Steel Pipe. ASTM A53, Type E or S, Grade B steel or ASTM 501. Weight schedules shall be as specified on the drawings.

c. Anchor Bolts. ASTM F1554, grade 36, without heads.

d. Unfinished Threaded Fasteners. Where not otherwise indicated, ASTM A 307, Grade A, regular low-carbon steel bolts and nuts of hexagonal design, hot-dipped galvanized.

e. Finished Threaded Fasteners. Stainless steel cap screws and heavy semi-finished nuts of hexagonal design for exterior connections, unless otherwise indicated.

5A.3 FABRICATION.

a. General.

i. After performing all fabrication and welding operations, remove all sharp edges and burrs that could cause injury. Properly finish surfaces of exposed items so as to be free of visible defects.

ii. Cut, drill, or punch holes perpendicular to metal surfaces. Do not flame cut holes.

b. Welding. Comply with AWS D1.1 Code for procedures, appearance, and quality of welds. Weld all work to produce assemblies free of warpage.

c. Galvanizing.

i. All steel members, plates, and assemblies shall be hot-dipped galvanized in accordance with ASTM A-123, unless otherwise specified.

ii. Prepare all structural steel items for galvanizing by solvent cleaning, hand and power tool cleaning, and/or sandblasting as required for permanent adhesion of galvanizing.

5A.4 INSTALLATION.

a. Install all work plumb, level, and square in accordance with the drawings.

b. Apply high zinc-dust-content paint for repair of galvanized surfaces damaged by welding. Paint shall conform to ASTM A-780.

DIVISION 6 – WOOD AND PLASTICS
SECTION 6A
ROUGH CARPENTRY

6A.1 DESCRIPTION OF WORK. The extent of exterior carpentry work is indicated on the drawings and by the provisions of this section.

6A.2 MATERIALS.

a. Lumber. Lumber shall be stress-rated and marked #2 structural grade, regardless of species. Sizes indicated are nominal. All lumber shall be dressed S4S, seasoned, and have 19 percent moisture content. Hand select all lumber pieces for straightness and freedom from defects.

b. Plywood. Plywood shall be all-veneer construction of sizes indicated on the drawings, and comply with APA- The Engineered Wood Association, grade designation: APA BC, Exterior, or better.

c. Lumber and Plywood Preservative Treatment.

i. All exterior lumber and plywood shall be preservative treated, and shall comply with AWPA Standard U1, Commodity Specification A: Sawn Products (Lumber) and Commodity Specification F: Composites (Plywood) to the requirements of AWPA Use Category UC3B. All material shall be quality marked in accordance with AWPA Standard U1 and shall bear the mark of an agency accredited by the American Lumber Standard Committee.

ii. Pressure treatments for lumber and plywood shall be with water-borne preservatives.

iii. Treat all cut surfaces and drilled holes in accordance with AWPA Standard M4

d. Fasteners. Provide type, size, and finish of fasteners indicated on the drawings. All exterior fasteners shall be hot-dipped galvanized or stainless steel.

6A.3 CONSTRUCTION.

a. Discard units of material with defects that could impair quality of work. Set carpentry work to required lines and levels with members plumb, level, and square. Accurately cut and fit all work.

b. Secure all carpentry work by anchoring or fastening as required by recognized standards. Make tight connections between members. Install all fasteners without splitting wood. Pre-drill as required.

c. Coat all exterior exposed cut edges and ends of lumber and plywood pieces with wood preservatives as required above.

DIVISION 13 – SPECIAL CONSTRUCTION
SECTION 13C
PAPI

13C.1 DESCRIPTION OF WORK. This section applies to special construction required for a Precision Approach Path Indicator (PAPI).

13C.2 FRANGIBLE SUPPORTS FOR PAPI EQUIPMENT.

a. Description. Frangible couplings will be used to support PAPI equipment installed near runways. Each frangible coupling has a hexagonal throat with a break-off groove in the middle, designed to break at low impact, thereby minimizing damage to colliding aircraft. The frangible point shall not be more than 3" above the finished grade.

b. Coupling Installation. When installing the frangible coupling, take care to use a wrench which will grip only the lower portion of the hexagonal throat of the coupling, i.e., that portion immediately below the break-off groove. If the wrench grips the upper portion of the hexagonal throat, the coupling may break when torque is applied. See Paragraph 16A.20 for thread remediation. Whether thread remediation is performed or not, apply anti-seize compound to the threads of the frangible coupling, to facilitate removal. The compound shall be an anti-seize assembly lubricant formulated to provide protection for stainless steel and dissimilar metal threaded fasteners against galling, seizure, and heat-freeze. Do not use plumber's pipe-joint compound. The frangible coupling shall be screwed down tightly into the conduit coupling.

13C.3 NOT REQUIRED

13C.4 ALIGNMENT AND AIMING ANGLE TOLERANCES.

a. Aiming Angles. Refer to site drawings for locations and aiming angles for individual PAPI light housing assemblies.

13C.5 SYSTEM CONTROL. Unless otherwise indicated PAPI system construction shall include a method of on/off control shown on the drawings or specified herein.

a. Air-to-Ground: the air-ground radio control receiver/controller gives the pilot for on-off operation of the PAPI.

b. Ground-to-Ground Radio Control System (RRCS): the RRSC system gives the ATCT controllers control of on-off operation of the PAPI. This system may also include an Air-ground radio for pilot control after ATCT stops daily operation.

c. Landline Control System: The landline control system is a hard-wired control system that gives the ATCT controllers (only) control of the on-off operation of the PAPI.

d. Fiber Optic Control System: The fiber optic control system is a hard-wired control system that gives the ATCT controllers (only) control of the on-off operation of the PAPI.

13C.6 OPERATIONAL TESTS. The Contractor shall demonstrate that the PAPI system will operate satisfactorily by a series of operational test cycles and a continuous test run of 24 hours minimum. The tests shall clearly indicate that the system meets all the requirements of the Drawings, Specifications, and the manufacturer's instruction manuals.

DIVISION 16 – ELECTRICAL
SECTION 16A
BASIC METHODS AND MATERIALS

16A.1 APPLICABLE DOCUMENTS.

a. Standards. The following standards in effect on the date of the invitation for bids or request for proposals, form a part of this Specification.

i. UL 6 : Electrical Rigid Metal Conduit – Steel

ii. UL 797 : Electrical metallic Tubing – Steel

iii. UL 360 : Standard for Liquid-Tight Flexible Steel Conduit

iv. FA-STD-019E : Lightning and Surge Protection, Grounding, Bonding and Shielding Requirements for Facilities and Electronic Equipment providing service.

v. UL 651 : PVC conduit.

vi. Local governing body rules and regulations.

16A.2 REQUIREMENTS.

a. General. The Contractor shall install all electrical work in accordance with the applicable Drawings and Specifications. All electrical work shall be installed to meet the provisions of the current issue of the National Electrical Code, NFPA-70, and all state and local regulations.

b. Contract Drawings.

i. Where the electrical drawings indicate (diagrammatically or otherwise) the work intended and the functions to be performed (even though some minor details are not shown), the Contractor shall furnish all equipment, material (other than Government-furnished items) and labor to complete the installation work, and accomplish all the indicated functions of the electrical installation.

ii. Minor departures from exact dimensions shown on the drawings may be permitted where required to avoid conflict or unnecessary difficulty in placement of the dimensioned item, provided all other contract requirements are met. The Contractor shall promptly obtain approval from the FAA Contracting Officer's Representative (COR) for any such proposed departure.

c. Materials. Materials and equipment, to be acceptable, must comply with all Contract requirements. Materials to be furnished by the Contractor under this Specification shall be new and, unless specified otherwise, the standard products of a manufacturer's latest designs. Wherever standards have been established by Underwriters Laboratories, Inc., the materials shall bear the UL label.

16A.3 WIRING METHODS.

a. General. All wiring shall consist of insulated copper conductors installed in metallic raceways, unless otherwise specified.

i. Conductor Routing. Panelboards, surge arresters, disconnect switches, etc., shall not be used as raceway for conductor routing other than conductors that originate or terminate in these enclosures. Isolated ground conductors will be allowed to traverse these enclosures.

ii. Conductor Separation. Power conductors shall be routed separately from all other conductor types. This may be accomplished by routing power conductors and other conductors in separate raceways, or by a metallic divider between the power conductors and the other conductors in the same raceway.

iii. Neutral Conductor: Shared/common neutrals shall not be permitted, i.e. each overcurrent device shall have its own separate neutral conductor. Neutral conductor sizes shall not be less than the respective feeder or phase conductor sizes.

b. Conductors.

i. Uninsulated Conductors: Uninsulated conductors shall be copper and in accordance with Military Specification A-A-59551.

ii. Insulated Conductors: Unless otherwise indicated, insulated conductors shall be copper with thermoplastic or thermosetting insulation, type THW, THWN and XHHW for general use or type THHN for use in dry locations only. All insulated for 600 volts in accordance with Federal Specification A-A-59544. Unless otherwise indicated, conductors #10 AWG and smaller shall be solid, conductors #8 AWG and larger shall be stranded. Minimum branch circuit conductor size shall be #12 AWG. Stranded conductors may be used with wire compression connectors or a pressure washer type lug; lugs with screw only compression are not allowed. Minimum control wire size shall be #14 AWG unless noted otherwise. Stranded conductors smaller than #10 AWG are allowed in applications where vibration and flexing may be encountered.

iii. Fixture Wiring: fixture wiring shall be thermoplastic insulated copper, rated for 600 volts, in accordance with federal Specification A-A-59544 and the NEC.

iv. Color Coding: All feeder and branch circuits, including neutral conductors, shall be identified at both ends of the conductor with panel and circuit number indicated. This shall be accomplished using shrink embossed labels only. The color coding shall be continuous throughout the facility on each phase conductor to its point of utilization so that the conductor phase connection is readily identifiable. Equipment grounding conductors shall be color coded green. Conductors covered with green insulation with yellow, orange, violet, or red tracers shall be used for other grounding systems. Neutral conductors shall be white insulated for 120/208/240 volt systems and gray insulated for 277/480 volt systems. For conductors #4 AWG and larger, where appropriate insulation color is not available, color coded tape, half lapped for a minimum length of 3 inches shall be used. Switch leg conductors shall be violet insulated. Green, white, and gray insulated conductors shall not be reidentified. All conductor color codes, including reidentified conductors, shall be visible at all junction boxes, pullboxes, panelboards, outlets, switches, access locations in closed raceways, every 3 feet in open raceways, under all raised floors, and at all terminations. Phase conductors shall be color coded as follows:

1. Single Phase

120 Volts	120/208/240 Volts
Line 1 – Black	Line 1 – Black
	Line 2 – Red
Neutral – White	Neutral - White

a. Color coding for conductors in control cables shall be in accordance with Standard ICEA S-61-402. DC power conductors shall be color coded as follows: positive conductor, red with brown tracer; negative conductor, brown with red tracer; neutral conductors, if used, shall be white.

16A.4 CONDUIT.

a. Where electrical metallic tubing is shown on the drawings, at exterior or interior locations, it shall be used without substitution.

b. Except where specified otherwise, conduit exposed to the weather, in concrete, or below grade shall be galvanized rigid steel with threaded joints. All conduit and conduit fittings in contact with earth shall have a factory PVC coating.

c. Except where otherwise specified, conduit used entirely indoors shall be rigid or electrical metallic tubing. Compression type fittings shall be used with metallic tubing.

d. Minimum size of conduit shall be 3/4-inch unless otherwise noted on the drawings. Each conduit run shall be installed complete before cable is pulled through.

e. All outdoor connections of conduit to enclosures shall be made with weatherproof hub fittings unless otherwise specified. Hub fittings shall have an external ground to maintain raceway grounding and be UL listed for grounding. Indoor connections of rigid conduit to enclosures shall be made with double locknuts and bushings. Refer to grounding section for disconnect switch conduit terminations.

f. Ends of conduits installed but not used, shall be closed with bushings and pennies. All underground conduits shall be temporarily plugged during construction to prevent entrance of foreign material.

g. Wherever conduit from outdoors or underground enters an enclosure or junction box, either indoors or outdoors, seals space between conduit and cables with conduit seal.

h. Exposed conduit shall be installed parallel to or at right angles with equipment and building wall surfaces unless shown otherwise. Field bends shall be avoided where possible, and where necessary shall be made with a hickey or conduit-bending device. Radius of field bends shall not be less than ten times the inside diameter of the conduit. Conduit shall be fastened securely to adjacent members or surfaces with galvanized clamps, straps.

i. Flexible metal conduit shall be used for terminal connections to motors or motor driven equipment, and in lengths only up to 6 feet for other applications permitted by the NEC. Liquid-tight flexible metal conduit shall be used outdoors and in wet locations. All flexible metal conduit shall be of a type where both the conduit and fittings are listed for grounding.

16A.5 GROUNDING.

a. Equipment, Structures, and Raceways.

i. All metallic non-current carrying parts of electrical equipment (including enclosures) and supporting structures installed under this contract, whether used either for power or control, shall be grounded with an equipment grounding conductor, whether or not shown on the drawings. The grounding conductor shall be sized in accordance with the National Electrical Code, but shall be of larger gauge if so shown on the drawings. In no case shall the grounding conductor be smaller than #12 AWG, unless shown otherwise on the drawings.

ii. A service entrance conduit or any other power feeder conduit emerging from below grade and supplying power to another facility or system component shall terminate with grounding bushings at both ends. These requirements apply unless shown otherwise on the drawings.

iii. The equipment grounding conductor shall be:

1. connected to the grounded conductor (neutral) only at the service entrance disconnecting means.

2. installed in the same conduit as its related branch and feeder conductors, and connected to the ground bus in the branch or distribution panelboard.

3. connected to all grounding bushings on conduits through which the conductor passes.

4. connected to all other grounding conductors in enclosures and bodies through which the conductor passes.

5. Each overcurrent device shall have its own equipment grounding conductor. No sharing of grounds between circuits is permitted.

iv. Where there are parallel feeders installed in more than one raceway, a properly sized equipment grounding conductor shall be installed in each raceway. The metallic conduit carrying the equipment grounding conductor shall be electrically continuous, forming a path parallel to the equipment grounding conductor. Under

no circumstances shall the equipment grounding conductor be omitted from the electrical system. Nor shall any separate grounding system such as the signal ground, be used for an alternate grounding system or an alternate path to the grounding electrode, unless so shown on the drawings.

v. All connections to the equipment to be grounded shall be made with a grounding connector specifically intended for that purpose. Connecting screws or mounting bolts and screws are not suitable for use as grounding connections. All ground lugs shall be of a non-corroding material suitable for use as a grounding connection, and must be compatible with the type of metal being grounded. Remove paint and other non-conducting materials from surfaces of grounding connections. After the connection is completed, apply touch-up paint or cold galvanizing to match existing.

vi. Unless otherwise specified, control equipment enclosures, pull boxes, and raceways, shall be grounded as above for power wiring.

vii. Where metallic conduit, surface-mounted square duct, or cable tray systems are installed, a separate copper conductor shall be installed in the raceway, and shall be properly bonded to each section. Unless otherwise specified, the minimum size ground conductor shall be #6 AWG green insulated copper.

b. Service Entrance Disconnect Switches and Breakers. All facility service entrance disconnect switches and breakers shall be grounded as follows:

i. The neutral bar or lug shall be grounded with a green insulated copper grounding electrode conductor, running directly to the grounding electrode. The grounding electrode conductor size shall be in accordance with the NEC, but in no case shall the wire size be smaller than #2 AWG.

ii. The switch box or panelboard enclosure shall be grounded to the grounded neutral bar or lug with a green insulated conductor, or other service grounding means.\

c. Grounding Electrode. Grounding electrodes (rods) shall be copper clad steel, 3/4-inch by 10 feet, except where otherwise specified. The top of the grounding electrode shall be a minimum of 12 inches below finished grade. Conductors shall be attached to electrodes with exothermic welds only, except where fire or explosion hazards exist, as near existing fuel tanks. Where such hazards exist, hydraulically crimped connections will be permitted as specified below.

d. Grounding Conductor. All grounding conductors shall be copper. All grounding conductors which are totally above grade shall be green-insulated conductors. All grounding conductors which are either entirely or partially direct-earth buried, shall be #6 AWG bare conductors, unless noted otherwise on the drawings.

e. Buried Guard Wire. A #1/0 AWG bare copper stranded guard wire shall be provided for all buried cables and conductors not routed in ferrous conduit. The guard wire shall be embedded in the soil a minimum of 10 inches above the cable to be protected and located directly above and parallel to the lines or cables being protected. When the width of the cable run or duct does not exceed 3 feet, one guard wire, centered over the cable run or duct, shall be installed. When the cable run or duct is more than 3 feet in width, two guard wires shall be installed. The guard wires shall be spaced at least 12 inches apart and be not less than 12 inches nor more than 18 inches inside the outermost wires or the edges of the duct. The guard wire shall be bonded to the earth electrode system (EES) at each end and to ground rods at approximately 90-foot intervals using exothermic welds. The spacing between ground rods must vary 10 percent to 20 percent to prevent resonance. Install the ground rods at approximately 6 feet on either side of the trench. Where cables run parallel to the edge of a runway, they shall be located 10 feet from the edge lights on the outside of the lights.

f. Exothermic Process for Connecting Grounding Conductors to Metal Objects. Where the Drawings and/or Specifications require connection of a grounding conductor to a metal object by exothermic process, the Contractor shall supply the correct exothermic welding kit for the application. The mold and cartridge used shall be selected on the basis of size, number, and type of conductors to be connected, composition and surface shape of object, and position in which the weld will be made. Regardless of the source of the kits selected, the Contractor shall submit catalog cuts or other manufacturer

information, demonstrating that the kits fit their intended applications on the above described basis. See Paragraph 1A.4 above. The Contractor shall provide and use the proper preparation tools in applying the exothermic process to ensure an adequate weld. Torch welds and/or brazing will not be permitted. No single-use exothermic weld molds, such as Thermoweld "Single Shot" and Cadweld "One Shot", will be permitted.

g. Hydraulically Crimped Connections. Hydraulically crimped connectors shall only be allowed where fire or explosion hazards prevent the use of exothermic welds. In such instances, grounding conductors shall be connected to each other or grounding electrodes with compression connectors crimped with a force of at least 24,000 pounds. Connectors, tools, dies, and crimping procedures shall be compatible to the application and to each other, and shall conform to the manufacturer's catalog and instructions. Each connector shall be clearly marked with catalog number, conductor size, and installation die information. The tooling shall be of the type that embosses or engraves the die index number on the connector in the crimping process. All connectors shall be listed in conformance with Underwriters Laboratories Standard UL467 and the National Electrical Code. Burndy Hy ground Compression System connectors, matching tools, and crimping procedures, are one system of products which meet these Specifications. Regardless of the source of the connectors, tools, and dies selected, the Contractor shall submit catalog cuts or other manufacturer information, demonstrating that these items fit their intended applications as described above. See Paragraph 1A.4 above.

h. Testing. The Contractor shall measure the resistance of the grounding electrode system in the presence of the RE. Tests shall not be conducted within 48 hours of a rainfall or in frozen soil. The resistance shall not exceed 10 ohms, unless otherwise indicated. If the measured resistance exceeds 10 ohms, the RE shall be notified immediately for further guidance. Upon project completion, the Contractor shall also submit a written test report to the RE. A tabulated report of the final resistance value at each location shall be provided in the report.

16A.6 GROUND FAULT INTERRUPTING RECEPTACLE. All outdoor receptacles provided by the Contractor shall be ground fault interrupting duplex receptacles in properly sized weatherproof boxes.

16A.7 CABLE ABANDONMENT. Ends of cables to be abandoned shall be buried two feet below grade unless otherwise specified. All above ground unused conduits, conductors, and cables shall be removed, unless the facility manager disapproves.

16A.8 WATERPROOFING CABLE ENDS. All cable ends which will be exposed to weather, water, ground, or corrosive environment prior to termination, shall be sealed against these elements while awaiting termination. This also applies to all cable ends in manholes or handholes. The sealing material shall be properly sized, easily removable heat shrinkable end caps, or electrical tape with an application of brushed-on protective electrical coating.

16A.9 ELECTRICAL EQUIPMENT NAME PLATES.

a. Each of the following types of equipment shall be identified with a name plate showing the functional name of the unit, voltage utilized, one or three phase as applicable, and additional information if specified or requested by the RE:

   i. Switches (except local lighting)

   ii. Panelboards

   iii. Main circuit breakers

   iv. Motor controllers

b. Name plates shall be non-ferrous metal or rigid plastic, stamped, embossed, or engraved with 3/8-inch minimum height letters and numerals. Name plates shall be secured to the equipment with at least two screws, except main breaker plates may be epoxy glued.

16A.10 PANELBOARD CIRCUIT DIRECTORIES. The Contractor shall clearly and neatly mark panelboard circuit directories, identifying each circuit established, re-established, or changed, as to the circuit's function.

16A.11 COVERING HOLES IN ENCLOSURES. No electrical enclosure will be accepted which has an unused open hole, except weep holes or vent holes. Holes in enclosures where conduits, bolts, or other objects were removed and not reinstalled, shall be closed with panels of the same material, thickness, color, and shade as the enclosure.

16A.12 SAFETY DISCONNECT SWITCHES AND FUSES. Safety disconnect switches and fuses shall meet the following Specifications.

a. General. Unless specified otherwise, all switches for circuit voltages of 600VAC or less, shall be heavy duty (Type HD), UL listed, and shall bear the UL label. The switches shall be NEMA 1, NEMA 3R, or NEMA 4X as required by the drawings or special Specifications.

b. Switch Interiors. All switches shall have switch blades which are fully visible in the OFF position when the switch door is open. All current-carrying parts shall be of high-conductivity copper, designed to carry the rated load without excessive heating. Switches shall have removable arc suppressors where necessary to permit easy access to line side lugs. Lugs shall be front removable and UL listed for 60°C or 75°C, aluminum or copper wires.

c. Switch Mechanism. Switches shall quick-make, quick-break, such that during normal operation of the switch, the operation of the contacts will not be capable of being restrained by the operating handle after the closing or opening action of the contacts has started. The operating handle shall be an integral part of the box, not of the cover. Switches shall have provisions for padlocking the switches in the OFF position with at least three locks. Switches shall have a dual cover interlock to prevent unauthorized opening of the switch door when the handle is in the ON position, and to prevent closing of the switch mechanism with the door open. The handle position shall indicate whether the switch is ON or OFF.

d. Enclosures. Covers on NEMA 1 enclosures shall be attached with pin type hinges. NEMA 3R enclosures shall be securable in the open position. NEMA 3R enclosures for switches through 200 amperes shall have provisions for interchangeable bolt-on hubs. Hubs shall accommodate the conduits of the diameters indicated on the drawings. NEMA 3R enclosures shall be manufactured from galvanized steel. All enclosures shall have a gray baked enamel finish, electrodeposited on cleaned, phosphatized steel.

e. Ratings. All fusible switches rated 100 through 600 amperes at 240 volts, and 30 through 600 amperes at 600 volts, shall have a UL-approved method of field conversion from standard Class H fuse spacing to Class J fuse spacing. The switch also must accept Class R fuses, and have provisions for field installation of a UL-listed rejection feature to reject all fuses except Class R. The UL-listed short circuit rating of the switches shall be 200,000 rms symmetrical amperes when Class R or Class J fuses are used with the appropriate rejection scheme. The UL-listed short circuit rating of the switch, when equipped with Class H fuses, shall be 10,000 rms symmetrical amperes.

f. Fuses. All fused switches meeting the above Specifications shall be fused with dual element, time-delay, UL Class RK5 fuses, of the continuous current rating specified on the drawings. One set of spares shall be furnished for each fusible device. The fuses' interrupting rating shall be at least 10,000 rms symmetrical amperes.

16A.13 PANELBOARDS AND CIRCUIT BREAKERS.

a. General. Panelboards shall be dead-front type, shall conform to Federal Specification W-P-115, Type I, Class 1, and shall be listed by UL except for installations which require special panelboards to incorporate items not available as UL listed. Panelboards shall be mounted so that the height to the top of the panelboard shall not exceed 81 inches above the finished floor level. Unless otherwise specified, panelboards shall have a full hinged front cover with a hinged door in that cover for access to circuit breaker switches. Doors shall have flush-type cylinder locks and catches. Doors over 48 inches in height shall have auxiliary fasteners on top and bottom. All locks in a project shall be keyed alike, and two keys shall be furnished with each lock. Directories shall be typewritten to indicate the load served by each circuit and shall be mounted on the inside of the door in a holder with a protective covering. Circuits shall be connected as indicated on the drawing. The directory shall

be arranged so that the typed entries simulate circuit breaker positions in the panelboard.

b. Wiring Gutters. The minimum size of side wiring gutters shall be 4 inches for power feeders up to and including 100 amperes, 6 inches for power feeders over 100 amperes and up to 225 amperes, and 8 inches for power feeders over 225 amperes and up to 600 amperes.

c. Circuit Breakers. All circuit breakers shall be UL listed thermal magnetic type or electronic solid state type, as described herein, and with a minimum rating of 10,000 AIC. Circuit breakers shall also have trip ratings, voltage ratings, and number of poles as defined on the drawings. All circuit breakers shall have a trip indicating feature. Single-pole breakers shall be full-size modules. Two-pole and three-pole breakers shall be physically sized in even multiples of a single-pole breaker. Breakers shall be sized so that two single-pole breakers cannot fit in a single housing. Multi-pole circuit breakers shall have an internal common trip mechanism. All circuit breakers and the panelboards in which the breakers are installed shall be products of the same manufacturer. Plug-in type load centers and/or plug-in type branch or feeder circuit breakers shall not be used.

   i. Thermal Magnetic. All thermal magnetic breakers shall be quick make, quick break type conforming to Federal Specification W-C-375. Adjustable breakers shall have setting adjustments readily accessible and visible from the front of the panelboard, after installation.

   ii. Solid State. Adjustable, solid-state or microprocessor-controlled circuit breakers shall have adjustments readily accessible and visible from the front of the panelboard, after installation. Individual circuit breaker frame size shall not exceed the panelboard bus rating.

d. Bussing Assembly and Temperature Rise. Panelboard bus structure and main lugs or main circuit breaker shall have current ratings as shown on the drawings. Such ratings shall be established by heat rise tests, conducted in accordance with UL Standard 67. Bus structures shall be insulated. All current carrying parts shall be of high-conductivity copper, designed to carry the rated load without excessive heating.

e. Cabinets and Fronts. The panelboard bus assembly shall be enclosed in a steel cabinet. The rigidity and gauge of the steel shall be as specified in UL Standard 50 for cabinets. Wiring gutter space shall be in accordance with UL Standard 67 for panelboards. The box shall be fabricated from galvanized steel or equivalent rust-resistant steel. Fronts shall have adjustable indicating trim clamps which shall be completely concealed when the doors are closed. Doors shall be mounted with completely concealed steel hinges. Fronts shall not be removable with the door in the locked position.

16A.14 ELECTRICAL ENCLOSURES AND WIREWAYS. Unless specified otherwise, electrical enclosures and wireways shall meet the following Specifications.

a. Material. Electrical enclosures and wireways shall be constructed of code gauge sheet steel.

b. Corrosion-Resistant Coating. Enclosure and wireway sheet steel shall be coated by ASTM 525 G90 (galvanneal) galvanizing or corrosion-resistant phosphate primer, or both.

c. Finish. Finish shall be dark gray enamel inside and out, or ANSI 61 gray polyester coating inside and out, or ASA-49 gray epoxy paint inside and out.

d. Industry Standards.

   i. Enclosures. NEMA 1 enclosures shall meet NEMA Type 1 and UL 50 Type 1 standards. NEMA 3R enclosures shall meet NEMA Type 3R and UL 50 Type 3R standards.

ii. Wireway. NEMA 1 wireway (including troughs) shall meet NEMA 1 and UL 870 standards. NEMA 3R wireway (including troughs) shall meet NEMA 3R and UL 870 standards.

e. Hardware. All hardware shall be plated to prevent corrosion.

16A.15 ELECTRICAL SURGE PROTECTION.

a. General. All electrical surge protection systems shall be installed in accordance with FAA Standards.

b. Supply Transformer. For utility owned transformers, protective devices shall be at the discretion of the utility. For FAA owned transformers, proper protection shall be provided on the primary side of the transformer.

c. Service Entrance Surge Arrester. The arrester shall be compatible with the service voltage, and shall be wired to avoid loops, sharp bends and kinks, and to minimize the number of bends. There shall be no interconnection between neutral and ground within the arrester. Arrester conductors shall be #4 AWG insulated copper or larger, unless a smaller size is recommended by the arrester manufacturer.

d. Transient Suppression Installations. Where transient suppression devices are installed in the electrical power distribution system, they shall be installed in accordance with the manufacturer's instructions unless otherwise specified.

e. Landline/cable Penetration Installations. Supression systems shall be provided for landline and cable penetration systems. High energy grounding conductors shall be bonded directly to the earth electrode system or to the perimeter ground cable under raised floors in equipment rooms with an approved fastener not more than 24 inches from the box.

16A.16 SHELTER LIGHTNING PROTECTION EQUIPMENT. All shelters (buildings) shall have a lightning protection system installed per the requirements of the Lightning Protection Code, National Fire Protection Association (NFPA 78), and Underwriters Laboratories Master Labeled System (UL96A). Lightning protection equipment items shall meet the following Specifictions:

a. Air Terminal Point. Air terminal points shall be nickel-tipped copper, 1/2" diameter x 36" long.

b. Point Bracket. For a roof ridge, the point bracket shall be made of pressed copper, shall bend to fit any roof slope, and shall hold the point and cable slightly above the center of the roof ridge. The bracket shall have a pressure cable clamp, and a stud to engage the point.

c. Air Terminal Brace. The air terminal brace shall be a 24"-long galvanized tripod assembly, with legs adjustable to accommodate any roof slope.

d. Roof and Down Conductors. Roof and down conductors shall each have 32 strands of #17 AWG copper wire, 7/16" overall diameter, braided smooth twist, 65,500 circular mils, and a net weight of 215 pounds per 1000 feet.

e. Ridge Cable Support. Ridge cable supports shall be pressed copper cable supports at least 2" wide, to hold the roof cable above the top of the roof. The ridge cable supports shall be sized to accommodate the roof conductor.

f. Cable Holder. Cable holders shall be 1"-wide copper bent-strap type loops with 1/4" mounting holes. The cable holders shall be sized to accommodate the roof conductor.

g. Parallel Clamp. Parallel clamps shall be bronze 2"-long clamps for connecting two conductors together, one conductor of maximum diameter 1/2", and the other conductor from 1/6" diameter to 5/16" diameter.

h. Flexible Bonding Strap. Flexible bonding straps, for connecting steel doors to steel door frames, shall be braids each composed of 480 #30 AWG copper wires, with flat bronze or

copper connectors crimped on at each end. The connectors shall have holes to take either 5/16" or 3/8" machine screws.

i. Pipe Clamp. Pipe clamps shall be adjustable tinned bronze clamps for bonding cables to pipes, and fitting pipes up to and including 1 1/4" O.D., and cables up to and including 1/2" diameter.

j. Bonding Equipment. Bond the steel siding, vent fan, hood, door frame, junction boxes, and any miscellaneous exterior metal objects to down conductors. If included, air conditioners, junction boxes, and flight check antenna masts shall be likewise bonded. Use the following equipment to perform the bonding:

i. Bonding Plate. Bonding plates shall be 8-sq. inch tinned bronze plates with 2"-long pressure type cable connectors, designed to bond a continuous run of cable to metallic objects along their path. Each plate shall have two holes fitting sheet metal screws or 1/4" machine screws.

ii. Bonding Conductor. Bonding conductor shall be minimum #6 AWG bare soft drawn copper, 1/6" diameter, 26,250 circular mils, net weight 80 pounds per 1000 feet.

16A.17 CONTROL CABLE TERMINAL STRIPS. Unless specified otherwise, Contractor furnished control (telephone) cable terminal strips shall be units assembled from compatible components all from the same manufacturer. The individual blocks of the strips shall be miniature style (6mm O.C.) nylon blocks with screw-activated tubular conductor clamps. The blocks shall be rated for a maximum voltage of at least 300 volts and a maximum current of at least 30 amperes. The conductor clamps shall accept wire sizes at least from #14 AWG to #22 AWG. Stab-in wire connection blocks shall not be used. The blocks shall be mounted in a standard 35mm DIN rail mounting channel. The assembled strip of blocks shall have a marking strip and holding plugs or end barriers. For terminating control cables on these strips, see Paragraph 16F.11 below.

16A.18 ELECTRICAL COATING. Cable connections, splices, or other joints wrapped with plastic electrical tape, shall be sealed with an electrical coating. Scotchkote Electrical Coating is among the products meeting this Specification.

16A.19 EXPANSION COUPLING. Where shown on the drawings, rigid metal conduits which emerge vertically from below grade to make a direct connection to an above-grade junction box or structure, shall be fitted with an expansion coupling. The purpose of the expansion coupling is to accommodate relative vertical movement, such as the movement due to frost heave. The coupling shall be rigid metal, and shall be threaded onto the rigid conduits at both ends of the coupling. The coupling must accommodate 8 inches of movement, unless space limitations prohibit installing such a coupling. If there are such space limitations, a coupling allowing only 4 inches of movement may be substituted. Electrical continuity across the expansion coupling must be maintained by installing a bonding jumper.

16A.20 SIGNAL AND COMMUNICATIONS.

a. Entrance Conduits. Conduit materials shall be rigid steel unless otherwise indicated. Except where otherwise indicated, underground conduits shall be a minimum of 2 feet below finished grade and extend at least 5 feet beyond the grounding electrode system. The conduits shall be bonded to the grounding electrode system with #2 AWG bare copper conductor by exothermic welds. Conduits installed for future use by others, such as for telephone, communications, electronic signals, etc., shall have both ends capped.

b. Transient Protection Demarcation Box for Electronic Landlines. A metallic, appropriately rated NEMA junction box, shall be installed where electronics landlines or conduits enter the facility. This box will house terminal boards, cables, and circuit transient protectors as shown on the Contract Drawings.

c. Fiber Optics. The use of fiber optics is recommended to replace metallic, control cables. Using fiber optics will eliminate outages and loss of service due to lightning strikes.

16A.21 REPAIR OF EXISTING WORK. Where cutting, channeling, chasing, or drilling of floors, all partitions, ceilings, or other surfaces is necessary for the proper installation, support, or anchorage of the conduit, raceways, or other electrical work, it shall be carefully done. The Contractor shall repair, with equal material by skilled workers, any damage to facilities caused by the Contractor's workers or equipment. The Contracting Officer's prior approval must be obtained for the materials, workers, time of day or night, repair method, and for temporary or permanent repair purposes. On completion, repair work shall be inspected and approved by the RE with the concurrence of any other affected parties such as utility companies and airport authorities.

16A.22 QUALITY ASSURANCE PROVISIONS.

a. List of Materials and Equipment. When required by the contract, the Contractor shall submit a list of materials and equipment to the Contracting Officer for approval.

b. Information Required. This list shall include manufacturer's style or catalog numbers. Partial lists submitted from time to time shall not be considered as fulfilling this requirement. Approval of materials will be based on manufacturer's published data. Approval of materials and equipment will be tentative, subject to submission of complete shop drawings, when required, indicating compliance with the Contract documents.

c. Statement. A manufacturer's statement indicating complete compliance with the applicable Federal Specification, Military Specification, or standards of ASTM, NEMA, or other commercial standard, is acceptable as indicating compliance with Contract documents.

d. Shop Drawings. When required by the Contract or by direction of the Contracting Officer, the Contractor shall submit shop drawings for materials and equipment not completely identified by information submitted in the materials and equipment lists. This information shall include, but is not limited to, panelboards, lighting fixtures, cable trays, switchgear, transformers, busways, cabinets, and lightning protection systems.

e. Coordination. Drawings and submitted data shall be checked and coordinated with the work of other construction trades involved, before they are submitted for approval, and shall bear the Contractor's stamp of approval as evidence of such checking and coordination.

f. Required Data. Drawings and submitted data shall be complete, assembled in sets and shall bear the date, drawing revision number, name of project or facility, name of Contractor and subcontractor, and the clear identity of contents and location of work.

g. Approval. The approval of drawings and submitted data shall not be construed as (1) permitting any departure from the Contract requirements; (2) relieving the Contractor of the responsibility for any errors, including details, dimensions, materials, etc.; or 3) approving departures from full size details furnished by the Contracting Officer.

h. Variations. If drawings show variations from the contract requirements because of standard shop practice or for other reasons, the Contractor shall describe such variations in a letter of transmittal to the Contracting officer. If acceptable, the Contracting Officer may approve any or all such variations, subject to a proper adjustment in the Contract. Contractors failing to describe such variations shall not be relieved of the responsibility for executing the work in accordance with the Contract, even though such drawings have been approved.

i. Submission. The Contractor shall submit and obtain approval of shop drawings by the Contracting Officer before ordering materials or proceeding with any work associated with the shop drawings.

16A.23 TESTS.

a. General. Unless otherwise indicated, the Contractor shall furnish all test instruments, materials and labor necessary to perform the following tests. All tests shall be performed

in the presence of the RE. All instruments shall have been calibrated within a period of 2 years preceding testing. Calibrations shall be traceable to applicable industry recognized standards.

b. Load Balancing. After the electrical installation has been completed, the Contractor shall take current readings with a true RMS ammeter for the purpose of load balancing. These readings shall be taken at the service entrance, each feeder panelboard, each branch panelboard, and each separately derived source. The Contractor shall redistribute single-phase loads where there is greater than a 20% difference between readings in any two phases. The Contractor shall also be required to notify the Contracting Officer of current readings taken before and after installation, and any phase loaded above 80% of the rating of its overcurrent protective device.

c. Neutral Isolation Tests. For all new installations, the neutral in the service entrance switch shall be tested for isolation from ground with an ohmmeter capable of reading greater than 20,000 ohms.

d. Earth Resistance Test. The Contractor shall comply with the grounding electrode system testing requirements of paragraph 16A.5.h.

e. Operating Test. After the interior wiring system installation is completed, and at such time as the Contracting Offer may direct, the Contractor shall conduct an operating test for approval. The equipment shall be demonstrated to operate in accordance with the requirements of this Specification. The test shall be performed in the presence of the RE.

DIVISION 16 – ELECTRICAL
SECTION 16B
600-VOLT POWER CABLE FOR UNDERGROUND INSTALLATION

16B.1 SCOPE. This section covers the material requirements for all Contractor-furnished single-conductor 600-volt power cable required for direct earth burial installation. Installation of power cable is covered in Section 16F.

16B.2 GENERAL REQUIREMENTS. Cable construction shall include copper single conductor and XLP (thermosetting crosslinked polyethylene) insulation. Cable shall be UL listed as Type USE or RHW or RHH for use in circuits not exceeding 600 volts at conductor temperatures of 90C for continuous normal operation, 130C for emergency overload conditions, and 250C for short circuit conditions. Cables shall be suitable for direct burial and above-grade installation in wet or dry locations.

16B.3 APPLICABLE SPECIFICATIONS.

a. Underwriters Laboratories Standard 854 for Service Entrance Cables.

b. Underwriters Laboratories Standard 44 for Thermoset-Insulated Wires and Cables.

c. ICEA Publication Number S-95-658, NEMA WC70 for Nonshielded 0-2kV Cables.

d. Federal Specification A-A-59544.

16B.4 CABLE CONSTRUCTION. Cable characteristics shall include the following materials and construction:

a. Conductors. Conductors shall be solid or Class B stranded annealed uncoated copper, per UL Standards 854 and 44.

b. Separator. A suitable separator over the conductor may be used at the option of the manufacturer.

c. Insulation. Each conductor shall be insulated with XLP (crosslinked polyethylene) complying with the physical and electrical requirements of UL Standard 854 for Type USE and UL Standard 44 for Types RHW and RHH and ICEA Publication Number S-95-658. The insulation shall be applied lightly to the conductor and shall be free-stripping.

16B.5 IDENTIFICATION. The cable shall be identified by surface marking indicating manufacturer's conductor size and metal, voltage rating, UL Symbol and type designation, and year of manufacture.

16B.6 TESTS. Cable shall be tested in accordance with requirements of UL Standard 854 for Type USE, UL Standard 44 for Types RHW and RHH, and ICEA Publication Number S-95-658.

16B.7 DATE OF MANUFACTURE. Year of manufacture of all cable shall be no earlier than one calendar year immediately preceding contract award date.

16B.8 PACKAGING. All cable shall be provided on wooden or steel reels, and ends of all cable shall be sealed to prevent entry of moisture. All reels shall identify type, length, and year

of manufacture of cable packaged on such reels. All such identification shall be clearly provided by the manufacturer.

DIVISION 16 – ELECTRICAL
SECTION 16C
600-VOLT ARMORED POWER CABLE

16C.1 DESCRIPTION. This section covers the material requirements for all Contractor furnished 600-volt 3-conductor armored power cable required for direct earth burial installation. Installation of power cable is covered in Section 16F.

16C.2 GENERAL REQUIREMENTS. Cable construction shall include three copper conductors with XLP (thermosetting crosslinked polyethylene) insulation, galvanized steel interlocking armor, and PVC jackets under and over armor. Cable shall be UL listed as type MC for use in circuits not exceeding 600 volts phase to phase at conductor temperatures of 90°C in dry locations, or 75°C in wet locations, 130°C for emergency overload conditions, and 250°C for short circuit conditions in wet or dry locations. Cables shall be designed and labeled for direct burial use.

16C.3 APPLICABLE SPECIFICATIONS. The following Specifications form a part of this Specification to the extent specified herein:

a. UL Standard 1569 for Metal-Clad Cables.

b. UL Standard 44 for Thermoset-Insulated Wires and Cables.

c. ICEA Publication Number S-95-658, NEMA WC70 for Nonshielded 0-2kV Cables.

d. IEEE 383 Type Tests of Class 1E Electric Cables and Field Splices and Connections for Nuclear Power Generating Stations.

16C.4 CABLE CONSTRUCTION. Cable construction shall include the following materials end construction:

a. Conductors. Class B stranded annealed uncoated copper per ICEA.

b. Separator. A suitable separator over the conductor may be used at the option of the manufacturer.

c. Insulation. XLP crosslinked polyethylene meeting the requirements of ICEA and Type XHHW requirements of UL 44. Average thickness of insulation shall be as specified in UL 44 for Type XHHW conductors and in ICEA. Minimum thickness at any point shall be not less than 90% of the specified average thickness.

d. Phase Identification. Insulated phase conductors shall be printed with the numeral "1", "2", and "3" on the surface of the insulation.

e. Assembly. Three phase conductors shall be cabled together with a Class B stranded, uncoated copper grounding conductor and suitable nonhygroscopic fillers to make round. Length of lay shall not exceed 35 times the phase conductor diameter. The grounding conductor shall comply with the requirements of UL Standard 1569. A suitable nonhygroscopic cable tape shall be applied over the assembly.

f. Inner PVC Jacket. PVC meeting the requirements of ICEA, Section 7 and the sunlight resistant requirements of UL 1569. Average jacket thickness shall be in accordance with UL 1569. Minimum thickness at any point shall be not less than 70 percent of the specified average thickness.

g. Armor. Galvanized steel interlocked armor shall be applied over the inner PVC jacket. Armor shall be in accordance with UL requirements for Type MC cable and Section 6 of ICEA.

h. Outer PVC Jacket. PVC meeting the requirements of ICEA, Section 7 and the sunlight resistant requirements of UL 1569. Average jacket thickness shall be in accordance with UL 1569. Minimum thickness at any point shall be not less than 70 percent of specified average thickness.

16C.5 TESTS. Conductors and completed cables shall be tested in accordance with UL requirements for Type MC cables having XHHW conductors.

16C.6 IDENTIFICATION. Cable shall be identified by surface marking indicating manufacturer's identification, conductor size and metal, voltage rating, UL symbol and type designation, year of manufacture, and "direct burial" designation.

16C.7 DATE OF MANUFACTURE. Year of manufacture of all cable shall be no earlier than one calendar year immediately preceding contract award date.

16C.8 PACKAGING. All cable shall be provided on wooden or steel reels, and ends of all cable shall be sealed to prevent entry of moisture. All reels shall identify type, length, and year of manufacture of cable packaged on such reels. All such identification shall be clearly provided by the manufacturer.

16C.9 SUBMITTALS. For the specific cable that the Contractor proposes to use, the Contractor shall submit the manufacturer's complete cable specifications, including compliance with all cable requirements, codes, and standards referenced herein, and a drawing showing cable construction details. Submit these items, and receive the Contracting Officer's approval before installing any cable specified herein. See Paragraph 1A.4 above.

DIVISION 16 – ELECTRICAL
SECTION 16E
CONTROL CABLE

16E.1 SCOPE. This section covers the material requirements for all Contractor furnished exterior standard and gopher-resistant filled control (telephone) cable to be installed as shown on the Drawings. Installation of control cables is covered in Section 16F.

16E.2 GENERAL REQUIREMENTS.

a. Definition. The term "control cable" used throughout these Specifications and on the project drawings is a general FAA term for cable used to transmit voice and control functions. The required cable is termed "telephone" or "telephone exchange" cable by the cable manufacturing industry.

b. Quality. All control (telephone) cables shall be the standard products of a single major cable manufacturer and shall be designed and manufactured according to the highest industry standards. All cables shall be free of any imperfection which could affect serviceability and design life.

16E.3 APPLICABLE SPECIFICATIONS.

a. United States Department of Agriculture, Rural Utilities Service (RUS), Specification REA Bulletin 1753F-205 (PE-39) for "Filled Telephone Cable" latest edition, including all addendums and attachments thereto, forms a part of these Specifications and is applicable in its entirety.

b. Certain requirements, specified herein, supplement the requirements of Specification REA Bulletin 1753F-205 (PE-39), and shall receive special attention by the cable manufacturer and Contractor.

c. Cable conforming to Specification REA Bulletin 1753F-208 (PE-89) is an acceptable substitute for PE-39 cable. For purposes of brevity, only the standard cable construction for PE-39 cable is specified herein.

16E.4 STANDARD CABLE CONSTRUCTION. Cable characteristics shall include the following materials and construction.

a. Conductors. #19 AWG solid annealed copper.

b. Conductor Insulation. Solid polypropylene or polyethylene color coded in accordance with telephone industry "standard" coding.

c. Twisted Pairs. Individual conductors twisted into pairs with varying lays to minimize crosstalk.

d. Forming of Cable Core. Cables having 25 pairs or less are assembled into a single cylindrical group. Cables having more than 25 pairs are assembled in units, each individually identified by color coded unit binders.

e. Filling Compound. Water resistant non-hardening compound to fill and seal all interstices between the conductor pairs.

f. Core Covering. Non-hygroscopic dielectric tape.

g. Flooding Compound. Water resistant and bonding compound to fill all voids between the core wrap and shield and between the shield and jacket.

h. Shield. Corrugated electrically continuous and longitudinally applied 0.008 inch coated aluminum or 0.005 inch copper.

i. Jacket. High molecular weight polyethylene or high-molecular weight ethylene copolymer.

16E.5 GOPHER-RESISTANT CABLE CONSTRUCTION.

a. General. If gopher-resistant cable is required by drawings or special Specifications, cable construction shall comply with all construction requirements for standard cable in

Subsection 16E.4 above (including conformance with RUS Specification REA Bulletin 1753F-205 (PE-39) except for Item h, "Shield", which shall comply with the following:

i. 0.010 inch copper.

ii. 0.006 inch copper/stainless steel/copper bimetallic alloy.

iii. 0.007 inch Alloy 194 for 6 pr #19 cable.

iv. 0.006 inch Alloy 194 for cables larger than 6 pr #19.

v. 0.008 inch coated aluminum with 0.006 inch coated steel.

16E.6 CABLE IDENTIFICATION. In accordance with Specification REA Bulletin 1753F-205 (PE-39), all cable shall have jacket printed at periodic intervals with the name of the manufacturer, manufacturer's standard designation, year of manufacture, number of pairs, conductor gauge, sequential length marks, and notation signifying compliance with the specification (if not clearly referenced in the manufacturer's submittals). In addition, the gopher-resistant shield shall be clearly identified.

16E.7 DATE OF MANUFACTURE. Year of manufacture of all cable shall be no earlier than one calendar year immediately preceding contract award date.

16E.8 PACKAGING. In accordance with Specification REA Bulletin 1753F-205 (PE-39), all cable shall be stored and shipped on reels affording the required protection. Thermal wrapping shall be provided and ends of all cables shall be capped against exposure to moisture. All reels shall be labeled by the manufacturer and shall bear the manufacturer's name, year of manufacture, REA cable designation, description of cable, actual shipping length, and identification referenced to tests of record as required herein.

16E.9 TESTS.

a. All project cable furnished shall satisfy all test requirements of Specification REA Bulletin 1753F-205 (PE-39). Records of all such tests shall be retained by the manufacturer, and shall be promptly made available to the Federal Aviation Administration upon request. All tests shall be specifically and clearly referenced to all reels of cable furnished.

b. Basic cable design, for all project cable furnished, shall have proven acceptable to RUS through "qualification testing" according to their specification.

c. Electrical tests, according to their specification, shall be performed on 100 percent of all project cable furnished.

d. Quality assurance (capability) tests, according to their specification, shall be performed on such periodic production basis so as to represent quality of all project cable furnished.

16E.10 SUBMITTALS. Prior to procuring any cable specified herein, the Contractor shall submit the following documents for the specific cable that the Contractor proposes to use, to the Contracting Officer, and receive written approval therefrom (see Paragraph 1A.4 above):

a. Manufacturer's complete cable specifications, including manufacturer's statement of compliance with REA Specification PE-39 or Pe-89.

b. Drawing showing cable construction details

DIVISION 16 – ELECTRICAL
SECTION 16F
CABLE INSTALLATION

16F.1 DESCRIPTION OF WORK. The extent of work is indicated on the drawings and by the provisions of this section. Included in this section are installation, splicing, and testing of power and control cables.

16F.2 GENERAL REQUIREMENTS.

a. Service Interruptions. Existing sources of power and control are indicated on the drawings. For circuits actively in use, the Contractor shall coordinate temporary interruptions of service with users and suppliers, the RE, and the airport management.

b. Precautions. The Contractor shall take all reasonable precautions to protect existing underground equipment and utilities such as fuel tanks, water lines, and buried control and power cables. All known FAA power and control cables leading to and from any operating facility will be marked in the field by the RE for the information of the Contractor before starting work in the general vicinity. The Contractor shall contact utility companies and the airport sponsor for the location of existing utility lines and airport sponsor cables. Thereafter, through the entire construction period, buried equipment and utilities shall be protected from damage. The Contractor shall immediately repair, with equal material, by skilled workmen, any underground cables damaged by contract workers, equipment, or work. Prior approval from the Contracting Officer shall be obtained for the materials, workers, time of day or night, methods of repairs, and for any temporary or permanent repairs the Contractor proposes to make. Upon completion, any repair work shall be inspected and approved by the RE with the concurrence of the affected utility company or airport sponsor.

c. Cable Protection.

i. All cable ends which will be exposed to weather, water, ground, or corrosive environment prior to termination, shall be sealed against these elements while awaiting termination or splicing. This requirement also applies to all cable ends in manholes or handholes. The sealing material shall be properly sized, easily removable heat shrinkable end caps, or electrical tape, with an application of brushed-on electrical coating.

ii. Cables shall not be bent at radii less than radii recommended by the manufacturer, or 10 times cable diameter (12 times diameter for armored cable), whichever is greater. Any cables damaged in any way by sharp bending shall be replaced.

iii. Special care should be taken when working with filled cables, especially when the temperature is below 35°F. This type of cable becomes more difficult to bend and work as the temperature decreases, and there is a possibility of cable damage at temperatures near 0°F.

d. 600-Volt Wire and Cable Color Coding.

i. All single conductor 600 volt wire and cable for 120/240 volt power circuits shall be color coded black for line 1, red for line 2, and white for the neutral.

ii. For conductor sizes smaller than #8 AWG, conductor insulation shall be color coded. For sizes #8 AWG and larger, and for armored power cable, colored tape shall be used to identify the conductors if insulation is not color identified.

iii. Conductors shall be color-coded in junction boxes, square duct, terminal boxes, or any other place accessible to view. In no case shall green be used for other than grounding, nor white for other than the system grounded (neutral) conductor.

e. Cable Lengths.

i. Wherever possible, cable shall be run in one piece, without splices, from connection to connection. The number of splices shall be minimized. If the job plans do not include a schedule for laying each reel of cable, the Contractor shall provide such a plan for approval to the Contracting Officer prior to installing any of the cable. The

plan shall be predicated on the use of the longest practicable lengths of cable to minimize splicing requirements.

1. When cable cutting is required, cable ends shall be effectively sealed against moisture immediately after cutting. The method of sealing shall be approved by the RE. Bands of a radius less than eight times the diameter for rubber-covered or plastic-covered cable, or twelve times the diameter for metallic armored cable shall not be made. Cable that has been kinked shall not be installed.
2. When unreeling, an observer shall be stationed at the reel to report any cable irregularities. Unless specifically stated in the plans, non-armored cable shall be used in duct and armored cable used for direct-earth burial. Non-armored coaxial and series lighting cable may be direct earth buried when not otherwise specified. Grounding conductors, where required, shall be #6 AWG bare copper wire, minimum.

f. Workmanship. All work shall be done by experienced personnel regularly engaged in this type of work. All cable splices shall be performed only by experienced and qualified cable splicers. Before any cable splices are made, the Contracting Officer may request a sample splice be made for approval.

16F.3 DIRECT-EARTH BURIAL CABLE INSTALLATION.

a. Installation Method. Direct-earth burial cables shall be installed either by the trench and backfill method or by the cable plowing method in accordance with all the requirements specified herein.

b. General Requirements.

i. Underground cables shall be installed in straight lines between terminating locations or points of directional change.

ii. Unless otherwise specified, cables shall be installed a minimum of 24 inches and 36 inches below finished grade on airport property and off airport lands, respectively. When cable is routed under railroad tracks, it shall be in rigid-steel conduit or concrete encased duct with the top of the duct not less than 42 inches below the base of the rail.

iii. The Contractor shall prepare a schedule for installing each reel of underground cable and shall submit it to the RE for approval before installing any cable. The plan shall be predicated on use of the longest practical lengths of cable, in order to minimize splicing.

iv. A cable loop of at least 3 feet shall be left on each and of every cable run, on at least one side of every splice, and at all points where cable is brought above ground. A 3-foot minimum surplus cable length shall be left on both sides of splices in handholes and light bases. The slack loop shall be installed with the same minimum depth requirements as the cable run. Where cable is brought above ground, enough additional slack cable shall be left to make the required connections.

c. Trench and Backfill Installation Method.

i. Comply with all trenching, backfilling, compaction, and restoration requirements in Division 2.

ii. Where turf is well established and sod can be removed, it shall be carefully stripped and properly stored. The Contractor shall excavate all trenches for direct earth burial cable as follows:

1. At the depth specified in Paragraph 16F.3.b(2) above.
2. To a width of not less than 6 inches for a single or multiple runs of power, or control and signal cable.
3. To a width and depth which will provide horizontal or vertical separation of power cables as specified in Paragraph 16F.3.f from other power cables of different voltage ratings, or from any power cable and any control or signal cable.

4. Unless otherwise specified, all cables in the same location and running in the same general direction shall be installed in the same trench. Walls of trenches shall be essentially vertical so that a minimum of shoulder surface is disturbed. The bottom surface of trenches shall be essentially smooth and free from coarse aggregate. Unless otherwise specified, trenches shall be opened only for the time required to install and inspect cables. The trench shall be closed in the same working day.

iii. Where rock is encountered, it shall be removed to a depth of 3 inches below the required cable depth, and shall be replaced with a bedding material of earth or same containing no mineral aggregate particles that would be retained on a 1/4 inch sieve. When a ledge is encountered, the RE shall be consulted regarding alternatives such as re-routing, transition to overhead lines, or installation in rigid steel conduit.

iv. The Contractor shall unreel the cable adjacent to or over the trench and manually place it in the trench. Do not pull the cable into the trench or drag it along the trench.

v. Where more than one cable is installed in the same trench, maintain separation as hereinafter specified. Multiple cables shall be installed in the same relative positions throughout the cable trench. Cables shall not be stacked, crossed or intertwined in any manner.

vi. After underground cable has been installed, the trench shall be backfilled. The first layer of backfill shall be 3 inches deep, loose measurement, and shall be either earth or natural sand containing no material aggregate particles that would be retained on a 1/4-inch sieve. This layer shall not be compacted. The second layer shall be 9 inches deep, loose measurement, and shall contain no particles that would remain on a 1-inch sieve. The remainder of the backfill shall be excavated or imported material and shall not contain stone aggregate larger than 4 inches maximum diameter. The second and subsequent layers shall be thoroughly tamped and compacted to at least the density of the adjacent undisturbed soil and to the satisfaction of the RE.

vii. Trenches shall not be excessively wet and shall not contain pools of water during backfilling operations. Trenches shall be completely backfilled and tamped level with the adjacent surface. If necessary to obtain the desired compaction, backfill material shall be moistened or aerated. When sod is to be placed over a trench, backfill shall be stopped at a depth equal to the thickness of the sod to be used. Any excess excavated material shall be removed in accordance with instructions from the RE.

d. Restoration. Where sod has been removed, it shall be replaced as soon as possible after the backfilling is completed. All areas disturbed by the trenching, storing of dirt, cable laying, pad construction, and other work shall be restored to the original condition. Restoration shall include any necessary grading, fertilizing, liming, seeding, sodding, sprigging, or mulching as required to restore the disturbed area to match the adjacent area. Where trenching cuts through paved area, the surface shall be properly backfilled and resurfaced with paving similar to the original paving. Resurfaced areas shall be level with original paving, free from cracks and capable of withstanding full traffic loads without settling or cracking. The Contractor shall be held responsible for maintaining all disturbed and restored surfaces until final acceptance.

e. Cable Plowing Method.

i. Vibratory cable plowing equipment, adequate for installation of the types of cables to be installed and for the depth required, may be used, provided that soil conditions are suitable, equipment is in good working order, and proper installation procedures are utilized.

ii. While cable is being plowed into place, one person in addition to the operator of the plowing vehicle shall be present to assure that the cables do not kink or bind tightly while entering the plow.

iii. If, during plowing operations, it appears that the soil contains sharp objects, rocks over 2 inches in diameter, or any other hazard to the cable, plowing shall be

discontinued, and the RE notified. The RE shall determine whether plowing will be allowed to continue, or whether another cable placement method shall be used.

iv. The slice left by the plow shall be closed by tamping or other approved method, after cable placement, to minimize the disturbance of the surface by the slice.

f. Cable Separation - Direct Burial.

i. Where new buried power cables cross over or under control or telephone cables, power cables shall be installed in a length of PVC duct extending two feet each side of the crossing. Minimum separation shall be twelve inches.

ii. Power cables of the same circuit may be laid together in the trench without separation, except as noted below. Series lighting cables may be considered being of the same circuit.

iii. Power cables, or the same or different circuits of less than 600 volts, may be laid together in the same trench without separation.

iv. All power cables, 5,000 volts and below, shall be separated from all control, telephone, and coaxial type cables by a minimum of 6 inches.

v. Power cables of more than 5,000 volts, shall be separated from all other cables by a minimum of 12 inches.

vi. Control and signal cables may be installed without separation from each other.

g. Cable Guard Wires. Guard wires shall be provided in accordance with the grounding requirements of Paragraph 16A.5.e above.

16F.4 CABLE INSTALLATION IN UNDERGROUND DUCTS AND CONDUIT.

a. General. All underground ducts shall be:

i. Rigid-steel conduit (heavy wall) conforming to UL 6

ii. Rigid non-metallic conduit (duct) conforming to UL 651 and NEMA TC 2.

b. Conduit Burial. Rigid-steel conduits may be direct earth buried. Rigid non-metallic conduits shall be concrete encased.

c. Duct Size, Material, and Installation. Ducts shall be of the size, material, and type indicated on the Drawings or Specifications. Standard precast spacers shall be used for duct support and alignment. Where no size is indicated on the Drawings or Specifications, the ducts shall not be less than 4 inches inside diameter. All duct lines shall be laid to slope toward handholes, manholes, and duct ends for drainage. Grades shall be at least 3 inches per 100 feet. On runs where it is not practicable to maintain the slope all one way, the duct line shall be sloped from the center in both directions toward manholes, handholes, or duct ends. Pockets or traps where moisture may accumulate shall be avoided.

d. Access Penetrations. Where a steel conduit penetrates a wall of a manhole or handhole, a grounding bushing shall be provided. These grounding bushings shall be connected to each other and to the earth ground system with #6 AWG bare copper conductors.

e. Mandrel Requirements. The Contractor shall mandrel each duct he installs and each existing duct in which he installs or replaces cable. An iron-shod mandrel, not more than 1/4-inch smaller than the bore of the duct, shall be pushed through each duct with jointed conduit rods. The mandrel shall have a leather or rubber gasket slightly larger than the duct hole.

f. Spare Ducts. All spare ducts installed by the Contractor shall be provided with #10 AWG copper-clad steel pull wires or polyolefin pull lines with a minimum tensile strength of 200 pounds. The open ends of the spare ducts shall be sealed with removable tapered plugs of a type recommended by the duct manufacturers. The plug shall be adapted to firmly secure the pull wire.

g. Duct Protection. All ducts shall be securely fastened in place during construction and progress of the work, and shall be plugged to prevent seepage of grout, water, or dirt. Any duct section having a defective joint shall not be installed. Trenching for ducts shall be in accordance with Paragraph 16F.3 of this Specification.

h. Ducts Encased in Concrete. Concrete-encased duct shall be installed so that the top of the concrete envelope or conduit is not less than 18 inches below the bottom of paving when installed under runways, taxiways, and other paved areas; and not less than 18 inches below finished grade when installed in unpaved areas. All concrete encased ducts shall be placed on a layer of concrete not less than 3 inches thick prior to its initial set. Where two or more ducts are encased in concrete, the Contractors shall space them not less than 1-1/2 inches apart (measured from outside wall to outside wall) using spacers applicable to the type of duct. As the duct laying progresses, concrete not less than 3 inches thick shall be placed around the sides and top of the duct bank. End bells or couplings shall be installed flush with the concrete encasement where required. Interlock spacers shall be used every 5 feet to insure a uniform spacing between ducts. All bottom spacers shall be secured to 1-inch by 3-inch boards to prevent sinking and overturning. All joints in adjacent ducts shall be staggered a minimum of 24 inches apart and shall be made completely waterproof prior to concreting.

i. Ducts Without Concrete Encasement. Underground ducts shall be installed so that the tops of all such ducts are at least 18 inches below finished grade. Underground ducts, except rigid steel conduit, shall not be installed under paved areas, roadways, railroad tracks, or ditches. Trenches for single-duct lines shall be not less than 6 inches nor more than 12 inches wide, and the trench for two or more ducts installed at the same level shall be proportionally wider. Trench bottoms for ducts without concrete encasement shall be made to conform accurately to grade to provide uniform support for the duct along its entire length. A 3-inch layer of bedding material shall be placed around the ducts. The bedding material shall contain no particles that would be retained on a 1-inch sieve. The bedding material shall be tamped until firm. When two or more ducts are installed in the same trench without concrete encasement, they shall be spaced not less than 2 inches apart (outside wall to outside wall) in a horizontal direction or not less than 6 inches apart (outside wall to outside wall) in a vertical direction.

j. Guard Wires. Ducts, whether encased or not encased, which contain cables that are not routed in ferrous conduit, shall be provided with a guard wire in accordance with the grounding provisions of Paragraph 16A.5.e above.

k. Installation Equipment. Major equipment items, required for installing cable in underground ducts, shall include the following:

   i. Power winch.

   ii. Cable feed-in tubing guide capable of producing a uniform and rigid 3 and 4-foot and greater radius bend, and having a nominal diameter equal to that of the ducts.

   iii. Single pulleys or sheaves providing a minimum cable bending radius (not overall sheave radius) of 10 times the largest cable diameter. Such sheaves shall be used for minor cable bends within "through cable" manholes and at feed-in manhole rims (if necessary). Sheaves shall have ball or roller bearings.

   iv. Adjustable gang pulleys with three or more pulleys capable of producing up to a 4-foot smooth cable bending radius. Each pulley shall have minimum cable bending radius of 10 times the largest cable diameter.

   v. Lubrication equipment to pre-lubricate ducts, cables at guide-in tubing, and cables at intermediate pull-through manholes.

   vi. Cable reel support equipment including stands, arbor, and braking mechanism.

   vii. Dynamometer for measuring pulling tensions.

   viii. Communications equipment.

   ix. Pulling ropes or cords having the following characteristics:

      1. A working strength at least equal to the maximum allowable cable tensions as specified herein. "Working strength" is normally 10 to 14 percent of published rope "breaking strength."

      2. Rope or cord shall be a twisted or braided synthetic fiber unaffected by water and having a low level of elongation under load. Material shall have a texture

non-injurious to plastic duct when pulled against bends. Wire rope, if proposed, shall have a smooth and rigidly adhering synthetic material covering.

3. All pulling ropes or cords shall have swivel devices at cable attachment ends.

x. Cable lubricant specifically manufactured for electrical and control (telephone) cables. Do not use soap lubricants or those containing soap which are harmful to polyethylene- sheathed cables.

xi. Cable pulling devices (secured to ends of cable as specified below).

l. Cable Pulling Devices.

i. Pulling devices for securing cable to pulling rope shall be factory-installed pulling eyes, field-installed pulling eyes, or basket weave cable grips. All shall be provided with integral or separate swivels.

ii. Factory-installed pulling eyes necessitate that each cable pulling segment be cut to length by the cable manufacturer. Greater tensions and longer pulling lengths can be used with factory pulling eyes for straight duct bank segments.

iii. Field-installed pulling eyes for control cable shall be a 4-crimp series, sized to the cable. Power cable pulling eyes shall be a type secured to conductors and approved by the RE.

iv. Basket weave cable pulling grips shall be carefully sized to the specific diameters of the cables to be installed. Use grips with a rotating eye feature for power and control (telephone) cables.

m. Duct Cleanout and Pre-Lubrication.

i. If any new or existing underground duct or conduit displays any evidence of contamination by soil or other foreign matter, such ducts or conduit shall be cleaned with a stiff bristle brush, swabbed, and flushed clean with water under pressure, before proceeding with cable pulling operations. Even a minor amount of soil or sediment in the bottom area of a duct will greatly increase the coefficient of friction and pulling tension required. With soil contamination, cable lubricant is of little value. Therefore, it is of utmost importance that conduit be cleaned prior to installation of cable.

ii. It is the Contractor's responsibility to determine whether ducts designated for occupancy should be cleaned. The Contractor shall assume complete responsibility for any difficulties or damage to the cable in placing cable in ducts.

iii. In addition to cable lubrication as specified elsewhere, all ducts to receive cables under this contract, shall be pre-lubricated using the same lubricant as for cables. Lubrication shall be thoroughly applied with applicators designed for this purpose. Lubrication on cable only, will rub off to a large degree, especially at duct bank offsets at manholes.

n. Setting Up Cable Reels and Apparatus.

i. The Contractor shall inspect cable reels for flange protrusions which could damage the cable sheath. Also, the Contractor shall inspect for any obstructions that could interfere with proper unwinding of the cable.

ii. Careful control shall be exercised in the movement of cable reels. Where it is necessary to roll a reel to a desired location, it shall be rolled in the direction indicated by the arrows painted on the reel flanges. The reel shall not be allowed to tilt. A substantial runway of heavy planks should be employed where uneven ground conditions exist that may cause the reel to tilt. Where it is necessary to move a reel of cable with heavy equipment, a cable reel sling or equivalent should be used.

iii. In conduit sections containing curves, the cable reel shall be set up at the manhole near the curve unless other conditions do not permit.

iv. Cable reels shall be set up on the same side of the manhole as the conduit section in which the cable is to be placed. The reel shall be made level and brought into proper alignment with the conduit section so that the cable may be passed from the

top of the reel in a long smooth bend at maximum radius into the duct without twisting and making more than a 90-degree bend. This is of utmost importance in handling filled type cable in temperature ranges of 35°F and lower. Under no circumstances shall the cable be pulled from the bottom of a reel.

v. It is essential that the cable reel be in proper alignment and level during the placing operation. Incorrect location of the reel will cause unnecessary binding which will result in uneven cable feed.

vi. Do not permit adjacent turns of cable on the reel to stick together and cause binding as the cable is payed off the reel. Feed the cable by rotating the reel manually.

vii. Other cable support equipment, such as pulleys, sheaves, and gang-pulley equipment shall be set up rigidly within intermediate manholes to smoothly guide cables to exiting ducts.

o. Attaching Pulling Grips. All pulling grips shall be stretched onto the cables such that the entire lengths of the grip woven material will exert tension on the cable, thereby distributing stress. If the end of any cable grip (furthest from the cable end) does not grip as tightly as the lead end, secure same to cable with a steel banding. Inspect cable grips frequently, and the first pull of control (telephone) cable in particular (in the first intermediate manhole), to ascertain that this requirement is fulfilled. If any uneven gripping is evident, banding will be required for all remaining cable installation of the applicable cable type and size.

p. Feeding and Pulling Cable.

i. All cable shall be installed using methods that will prevent excessive and harmful stretching, twisting, and flexing of the cable. Such damaging treatment will mechanically weaken the cable and destroy the electrical properties immediately or in a short time.

ii. Cable may be pulled by hand or power winch. Pull rope shall be attached to cables with pulling eye or basket weave pulling grips (all equipped with swivels) for each cable pulled. Do not exceed maximum allowable pulling tension as hereinafter specified. Do not use cable manufacturer's maximum pulling tensions except for cable factory-installed pulling eyes.

iii. All splices shall occur in manholes only. Splices shall not be pulled into ducts or manholes.

iv. Cable feed-in tubing guide, same size as conduit, of suitable length shall be secured in the manhole between the cable reel and the face of the duct to protect the cable and guide it at the maximum possible smooth radius into the duct as it is payed off the reel.

v. A cable lubricator (funnel) shall be placed around the cable just ahead of the cable feed-in guide to facilitate lubrication of the cable. The quantity of lubricant shall conform to the lubricant manufacturer's recommendations.

vi. Before starting to pull, check the equipment carefully to make sure that it is properly set up in order to minimize the chance of interruption once pulling has started. Tension shall be kept on both the cable reel and the pulling line at the start of the pull. Excessive slack and the twist of the pulling line may cause the connecting links to turn and catch in the duct. As far as possible, the cable shall be pulled in without stopping. A pulling speed of 80 to 100 feet per minute is recommended to minimize friction forces.

vii. A person experienced with cable handling shall be posted continuously at the cable reel while pulling cable. In addition to braking the reels and observing cable lubrication, he shall carefully inspect cable paying off the reel for cable sheath and other defects. If defects are noticed, the pulling operation shall be stopped immediately and the RE promptly notified of the defect. Kinks and/or irregularities in the cable sheath shall be removed or corrected as directed by the RE.

viii. Careful attention shall be paid to signals from the installation crew as the cable is being pulled so that pulling may be stopped instantly whenever necessary to avoid damage to the cable.

ix. If for any reason the pulling operation is halted between manholes, the winch operator shall not release the tension on the winch unless directed to do so. In restarting the pulling operation, the inertia of the cable shall be overcome by gradually increasing the tension in steps a few seconds apart until the cable once again is in motion.

x. The leading end of the cable at intermediate manholes shall be guided into the duct and a feeder tube nozzle placed around the cable to prevent the cable from rubbing on the edge of the duct.

xi. All pulled ends shall be examined for evidence of damage due to the pulling operation. The cable sheath shall not be pulled beyond the cable core. Notify the RE for inspection, and for repair or replacement action that must be taken where cracks or openings are found in the cable sheath following the pulling operations.

xii. Cable ends shall be kept sealed at all times using RUS approved cable end caps and electrical tape. After the cable has been placed, the exposed cable in the manholes should be wiped clean of cable lubricant with a cloth before leaving the manhole.

xiii. All individual cable segments shall be pulled in one direction only. Both ends of a cut cable segment shall not be introduced into an intermediate manhole and pulled in two different directions. Also, no cable segments shall be pulled out of any manhole and introduced into the same manhole for a continuation of a cable segment pull. These unacceptable pulling practices, used to avoid splicing, result in abrasion from dragging over ground surfaces and manhole frame, exposure to pedestrian and vehicular traffic, damage to cable layers from twisting and small bending radii when pulling cable loops through manhole frame. Shields of cables so pulled are almost always damaged.

xiv. Sidewall cable pressure from duct bends, feed-in tubes, and pulleys, frequently govern the length of cable that can be pulled. The greater the radii, the less the sidewall pressure. Therefore, the Contractor shall use the maximum radius at every manhole where a 90-degree pull is permitted. Adjustable gang pulleys with three or more pulleys shall be used for horizontal bends in manholes. Individual pulleys within the gang pulley device shall have a cable bending radius of minimum 10 times outside diameter of largest cable to be pulled. Width of pulleys shall be adequate to support the cable group to be pulled. Adjust gang pulleys to produce a smooth 90 degree curvature bend where such changes in direction occur.

xv. If cables will be spliced in a manhole where duct banks enter and leave 90 degrees apart, separate cable segments shall be introduced into the manhole and pulled in different directions unless pulling is permitted around a horizontal gang pulley within the manhole.

xvi. Where more than one cable will be installed in a single duct, all shall be pulled into the duct concurrently.

q. Cable Spoil. All cable pulling ends shall be trimmed back to remove cable material always damaged by pulling eyes or basket weave pulling grips. To remove such spoil, cut each cable off a distance from the end equal to three times the length of pulling eye or twice the length of the basket weave pulling grip as a minimum. These amounts shall be cut off for all cables including those to be spliced or terminated by others.

r. Use of Dynamometer.

i. The dynamometer shall be accurately calibrated and secured to properly indicate tension exerted on the cable. The dynamo-meter reading will usually give the resultant force exerted on the anchoring device, which shall be converted to the horizontal component to give correct value of pulling tension.

ii. Dynamometer readings shall be made only in the presence of the RE. If any pulling tension is approaching the maximum allowable, and if in the judgment of the RE, the allowable will be appreciably exceeded for the proposed run, pulling operations shall be immediately stopped, and the cable run spliced in the preceding manhole.

s. Maximum Cable Pulling Tensions. Maximum allowable cable pulling tensions, as measured by dynamometer, shall not exceed the following values for single cables. For multiple cables, add the tension values for the number of cables being pulled. Use a

pulling rope having a working strength [not breaking strength, reference subsection 16F.4k(9)] at least equal to the "maximum allowable pulling tension" values below.

Cable	**Maximum Allowable Pulling Tension (lbs)**
1-1/C #8	125
1-1/C #6	200
1-1/C #4	325
1-1/C #2	500
1-6 PR #19	125
1-12 PR #19	250
1-25 PR #19	500

t. Separation of Cables Installed in Conduit or Duct.

i. Power cables of the same voltage may be installed in the same duct.

ii. Power cables of less than 600 volts may be installed in the same duct.

iii. Power cables of less than 600 volts shall not be installed in the same duct with control, telephone, or coaxial type cables.

iv. Power cables of more than 600 volts shall not be installed in the same duct with control, telephone, coaxial, or power cables of less than 600 volts.

v. Control, telephone, and coaxial cables may be installed in the same duct.

vi. Power cables may be installed in the same duct system as control and signal cables, but power cable shall be installed in a different duct separated a minimum of 3 inches (outside wall to outside wall) from ducts that encase control and signal cables. Power cables rated more than 600 volts shall be separated from control and signal cables to the maximum extent possible in the duct system.

u. Cable Installation in Manholes or Handholes.

i. Power and control cables shall be installed in separate manholes or handholes unless otherwise specified. If installed in same manhole, install power and control cables on opposite sides. At splice locations, use cable racks at different elevations to separate power and control cables.

ii. Cable racking surplus shall be pulled back by hand into intermediate manholes. Pull surplus one manhole at a time beginning near both ends of cable segment. Do not use power winch unless permitted by the RE.

iii. Cables shall be carefully routed around manhole interiors, taking all necessary precautions to prevent sharp bending. Cable racks shall be plastic or galvanized steel with properly sized porcelain insulators for the latter. Fasten all cables to plastic racks with nylon ties and to steel racks by means of the insulators.

iv. Where a splice occurs, cable shall make one loop around , and the splice located near the center of the loop.

v. Where power and control cables are installed in the same manhole, the entire exposed length of all power and control (telephone) cables shall be fireproofed by applying fire and arc proofing tape.

16F.5 CABLE LABELING.

a. All cables shall be labeled in each manhole and in each terminal cabinet with not less than two labels per cable, one near each duct entrance hole. Labels shall be attached to cables immediately after installation of each cable.

b. Labels shall be military grade, pre-cut, heat-shrinkable flattened polyolefin and shall readily accept thermal transfer print. The product shall meet the material requirements and physical properties of AMS-DTL-23053/5C (Class 1 and Class 3). Label size shall

be consistent with the manufacturer's maximum size for a particular cable size range, have a white background with 1/4-inch high black lettering, or the maximum size letters for labels that cannot accommodate the 1/4-inch height.

c. Labels shall consist of an abbreviation of the facility served , e.g., "ASR", "VORTAC", "ALS", and abbreviations for all the types of cables installed, e.g. "P" for power, "C" for control, and "R" for radio frequency (coaxial). Where like multiple control cables are routed between the same facilities, further identify such cables throughout the run with a single-digit number following the letter "C". All individual-conductor power circuits shall be bundled under the same label as opposed to separate labels for each conductor.

16F.6 WARNING TAPE. Warning tape shall be placed 12 inches above cables that are installed by direct burial, and ducts both with and without concrete encasement. The tape shall be detectable warning tape, 3 inches wide, and yellow in color with the words "CAUTION BURIED CABLE LINE BELOW" printed on it.

16F.7 CABLE MARKERS. The location of direct earth-burial cables, and ducts both with and without concrete encasement, shall be marked at grade surface with concrete slabs, 2 feet square by 6 inches thick. These markers shall be placed every 200 feet along a cable run, at each change of direction of the cable, and at each cable splice. These markers shall be installed within 24 hours of the final backfill of the cable trench. Markers shall be installed flat in the ground with the top approximately 1 inch above the finished grade. After the concrete marker has set a minimum of 24 hours, the top surface shall be painted with bright orange paint manufactured specifically for uncured exterior concrete. Markers shall not be installed in concrete or asphalt surfaces. Each cable marker shall have the following information impressed upon its top surface:

a. The word, "CABLE."

b. Name of facility served, e.g., "ASR," "VORTAC," "ALS."

c. Abbreviations for all the types of cables installed, e.g., "P" for Power, "C" for Control, and "R" for Radio Frequency (coaxial).

d. An arrow to indicate the direction or change of direction of the cable run.

e. Any additional information, as directed by the Contracting Officer.

f. Manholes and handholes shall be identified by "FAA POWER" or "FAA-CONTROL" markings on the covers. These markings shall be cast into steel covers, or die-stamped into a nominal 1/16 inch minimum thickness copper place razed or fastened to the cover with a minimum of two 10-32 brass screws. The Contractor shall obtain approval from the Contracting Officer for the information to be impressed on the cable markers and the method used to make the impressions. Letters shall be 4 inches high, 3 inches wide, and 1/2 inch deep.

16F.8 SPLICING.

a. General Requirements.

i. Splices shall be performed only by experienced and qualified cable splicers regularly engaged in this type of work.

ii. Cable armor and/or shielding shall be bonded together across splices to provide continuous electrical paths.

iii. Where a cable is cut preparatory to splicing, the work shall proceed without delay. When an unavoidable delay is encountered in completing a splice, the opened cable shall be protected to prevent the entrance of moisture and foreign matter.

iv. Any splicing material (such as resin) older than the do-not-use-after date on the package, shall be replaced with new material at the Contractor's expense.

v. Unless otherwise specified, where multiple runs of single-conductor underground power cables are spliced, each single-conductor cable shall be spliced in a separate envelope.

vi. Approved stress reduction methods shall be used in splicing all shielded high voltage power cables (5KV and higher voltage).

b. Underground Power Cable Splices (600 Volts or Less). All low voltage splices shall be encapsulated in pressure resin in clear plastic envelopes, except as otherwise specified in Paragraph 13A.6 above, on drawings, or in special Specifications. All low voltage splices shall be made with compression connectors specified in Paragraph 13A.6 above, except as otherwise specified on drawings and in special Specifications.

c. Underground Control (Telephone Cable) Splices.

i. Kit and Resin. The splices shall consist of a rigid polypropylene mold body with a built-in spacer web to provide cable centering and proper compound coverage. The mold body shall be filled with a flexible polyurethane electrical compound capable of continuous operation at 90°C, with an emergency overload temperature rating of 130°C. Splices must have provisions for inline splicing of shielded or non-shielded plastic or rubber-jacketed control (telephone) cables. The splices shall be rated for direct burial applications. For control cables with outside diameters between 0.25 inches and 3.25 inches, 3M Scotchcast Signal and Control Cable Inline Splicing kits of the 72N series are approved, as they are among kits which meet Specifications.

ii. Connectors. Control cable splice connectors shall be in-line type, in which two conductors are spliced by laying one conductor in each end of the connector, and crimping the connector with a special tool selected to match the connector type and size. Before crimping,the connector is open on one side of its length. After crimping, the connector is closed all around its length. The connector bodies shall be made with a tin-plated phosphor bronze piece on the inside, to contact the cable conductors, and bonded polyester insulation on the outside, to insulate the connection. The insulation shall be color coded to denote wire size range. The cable splice connectors and tools shall incorporate the insulation displacement termination technique which uses a slotted, tin-plated contact to displace the conductor insulation, thus providing four redundant electrical contact points. Connectors which require pre-stripping the conductor shall not be used. AMP, Inc. (Harrisburg, PA) Picabond connectors sized for conductor size, and matching AMP tooling, are among products meeting the above Specifications, and are approved.

d. Submittals. See Paragraph 1A.4 above. If the Contractor

i. Intends to splice using materials different from those specified in Paragraphs b and c above

ii. Intends to splice a 5KV or higher voltage power cable, then the Contractor shall submit to the Contracting Officer, shop drawings or catalog cuts for all splicing materials, tools, and dies. The Contractor shall splice no cables before he has received the Contracting Officer's approval of these items.

16F.9 CONTROL (TELEPHONE) CABLE TERMINATIONS.

a. Cable Routing and Support.

i. Cable jackets shall be removed within terminating enclosures such that no more than 2 inches of jacket material is visible within the enclosures. Ground shielding and armor as specified below.

ii. Exposed cable conductor bundles shall be lock-stitched laced together with nylon lacing twine spaced at approximate 5/8- inch intervals. Each bundle shall contain maximum 25 pairs of conductors which shall be neatly routed and secured to backing panels with nylon clamps.

b. Cable Pair Terminations.

i. Terminated pairs shall have the same sequence on each terminal strip. (For terminal block Specifications, see Paragraph 16A.19, above.) The color code termination sequence on the terminal strips shall be in accordance with the following schedule. The white mates shall start at the top or left-hand side of the terminal block with color continuing down or across the block according to the following schedule:

MATE COLORS	PRIMARY WIRE COLORS
WHTE	BLUE
	ORANGE
	GREEN
	BROWN
	SLATE
RED	BLUE
	ORANGE
	GREEN
	BROWN
	SLATE
BLACK	BLUE
	ORANGE
	GREEN
	BROWN
	SLATE
YELLOW	BLUE
	ORANGE
	GREEN
	BROWN
	SLATE
VIOLET	BLUE
	ORANGE
	GREEN
	BROWN
	SLATE

ii. When cables do not have the preceding color code, like pairs shall be terminated in the same sequence at both ends of the cable.

16F.10 CABLE ARMOR AND SHIELD GROUNDING.

a. Grounding Locations.

i. Control cable armor and/or shielding shall be grounded at both ends of each cable run. Ground unused pairs of control cable at both ends.

ii. Power cable armor shall be grounded at both ends of each cable run.

iii. Shielding and armor of control and power cables shall not be grounded at splice locations.

b. Grounding Procedures.

i. Use #16 AWG stranded copper grounding conductors for grounding shielding and armor. Secure grounding conductors to shielding and armor by using UL-approved grounding connectors specifically designed for this purpose. Neatly tape ends of butted cable to conceal the connections.

ii. Attach crimp-type lugs of proper size to free ends of grounding conductors, and secure lugs to enclosure interior wall with a machine screw and nut.

DIVISION 16 – ELECTRICAL
SECTION 16G
CABLE TESTING

16G.1 DESCRIPTION OF WORK. The extent of work is indicated on the drawings and by the provisions of this section. Included in this section are testing of power, control, and coaxial cables.

16G.2 GENERAL REQUIREMENTS.

a. Both before and after installation, all Contractor-furnished and Government-furnished power and control (telephone) cables shall be tested as required herein. Testing after installation shall be accomplished across splices.

b. All testing shall be accomplished in the presence of the RE. Furnish two signed and dated copies of all test resultsto the RE using the Technical Reference Data Record (TRDF) form in Appendix 4.

c. The Contractor shall use his own test equipment, which shall bear current calibration certification from a certified instrument calibration laboratory.

d. Any measured values not conforming to specified values shall be cause for rejection of the defective cable installation. After repair or replacement, if so required by the RE, cable shall be retested and additional remedial work performed until satisfactory test results are obtained. All repair and replacement work shall be accomplished at no additional cost to the Government.

16G.3 INSULATED POWER CABLES, 600 V. Insulation–resistance measurements must be made with both ends of the insulated conductors disconnected from source and load terminals, arresters, and large surge protectors. Apply a 500-volt dc voltage to the conductors being tested with an insulation-resistance test instrument and read the insulation directly on the instrument meter scale. Apply the test voltage for at least 1 minute after the meter reading has stabilized, to allow even distribution of the test voltage throughout the insulation. Apply the test voltage between each insulated conductor and all other conductors, and between each insulated conductor and ground. Record the measured resistance. It shall be a minimum of 50 megaohms.

16G.4 CONTROL CABLE TESTING. For continuity, shorts, crosses, and grounds testing, make these tests with both ends of each conductor disconnected from terminal strip jumper wires, surge protectors, and equipment. The tests should be as follows:

a. Continuity Tests. Make continuity tests with an ohmmeter at the near end of the cable. Temporarily short each conductor to its paired conductor, common connector, or to a shield or armor, at the far end of the cable. Record the meter readings. Remove the temporary short at the far end of the cable after each conductor or conductor pair is tested for continuity.

b. Shorts, Crosses, and Grounds Tests. Make the tests with an ohmmeter connected at either end of a cable. Test between all paired or unpaired conductors within a shielded or unshielded conductor group. Test between each conductor and its nearer designated grounding conductor, pair shield, group shield, cable shield, or armor.

c. For insulation–resistance measurements, apply a 500-volt dc test voltage to the conductors being tested with an insulation-resistance test instrument, and read the insulation resistance directly on the instrument meter scale. Apply the test voltage for at least 1 minute after the meter has stabilized. Apply the test voltage between paired conductors, between conductors of a conductor group, and between conductors of an ungrouped cable. Apply the test voltage between each conductor and its designated common grounded conductor, pair-shield, group shield, and overall cable shield or armor. The resistance shall be a minimum of 50 megaohms.

1.EOSH SAFETY ADVISORIES